and the spring was uncompressed. Therefore the wheel is 4 in. $\left(\frac{1}{3}\text{ ft}\right)$ below the equilibrium position (with respect to the motorcycle frame), and we have $x(0) = \frac{1}{3}$. According to the problem statement, the motorcycle has a velocity of 10 ft/sec downward when the motorcycle contacts the ground, so the spring will be given an equal and opposite velocity of 10 ft/sec upward, giving us $x'(0) = -10$. Applying these initial conditions, we get $c_1 = -\frac{3}{2}$ and $c_2 = \frac{11}{6}$, so the equation of motion is

$$x(t) = -\frac{3}{2}e^{-8t} + \frac{11}{6}e^{-12t}.$$

3. The graph is shown in Figure 17.3.10.

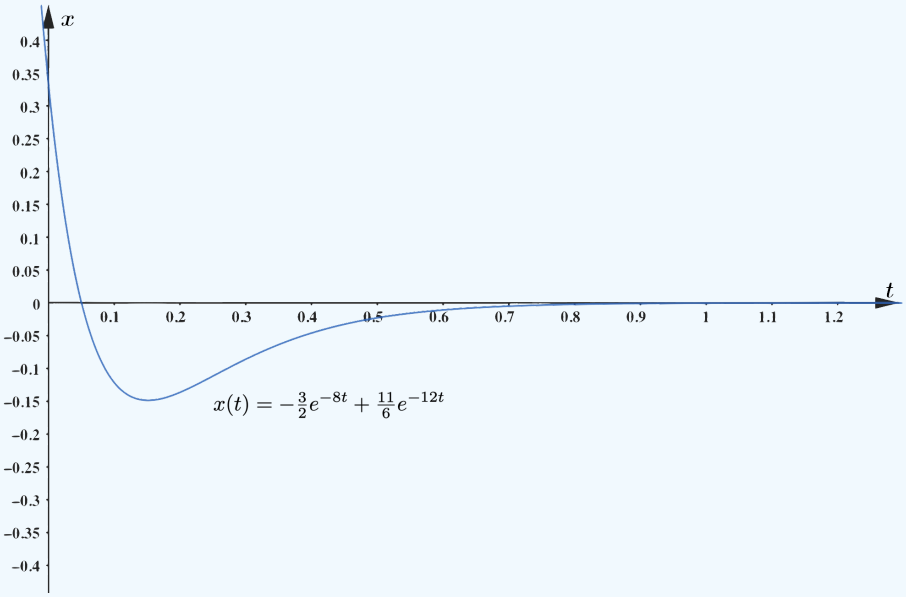


Figure 17.3.10: Graph of the equation of motion over a time of one second.

LANDING VEHICLE

NASA is planning a mission to Mars. To save money, engineers have decided to adapt one of the moon landing vehicles for the new mission. However, they are concerned about how the different gravitational forces will affect the suspension system that cushions the craft when it touches down. The acceleration resulting from gravity on the moon is 1.6 m/sec^2, whereas on Mars it is 3.7 m/sec^2.

The suspension system on the craft can be modeled as a damped spring-mass system. In this case, the spring is below the moon lander, so the spring is slightly compressed at equilibrium, as shown in Figure 17.3.11.

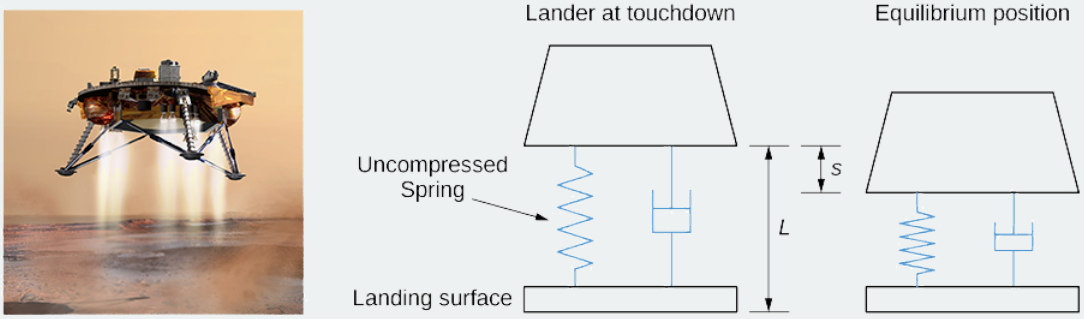


Figure 17.3.11: The landing craft suspension can be represented as a damped spring-mass system. (credit “lander”: NASA)

We retain the convention that down is positive. Despite the new orientation, an examination of the forces affecting the lander shows that the same differential equation can be used to model the position of the landing craft relative to equilibrium:

$$mx'' + bx' + kx = 0,$$

where m is the mass of the lander, b is the damping coefficient, and k is the spring constant.

1. The lander has a mass of 15,000 kg and the spring is 2 m long when uncompressed. The lander is designed to compress the spring 0.5 m to reach the equilibrium position under lunar gravity. The dashpot imparts a damping force equal to 48,000 times the instantaneous velocity of the lander. Set up the differential equation that models the motion of the lander when the craft lands on the moon.
2. Let time $t = 0$ denote the instant the lander touches down. The rate of descent of the lander can be controlled by the crew, so that it is descending at a rate of 2 m/sec when it touches down. Find the equation of motion of the lander on the moon.
3. If the lander is traveling too fast when it touches down, it could fully compress the spring and "bottom out." Bottoming out could damage the landing craft and must be avoided at all costs. Graph the equation of motion found in part 2. If the spring is 0.5 m long when fully compressed, will the lander be in danger of bottoming out?
4. Assuming NASA engineers make no adjustments to the spring or the damper, how far does the lander compress the spring to reach the equilibrium position under Martian gravity?
5. If the lander crew uses the same procedures on Mars as on the moon, and keeps the rate of descent to 2 m/sec, will the lander bottom out when it lands on Mars?
6. What adjustments, if any, should the NASA engineers make to use the lander safely on Mars?

Forced Vibrations

The last case we consider is when an external force acts on the system. In the case of the motorcycle suspension system, for example, the bumps in the road act as an external force acting on the system. Another example is a spring hanging from a support; if the support is set in motion, that motion would be considered an external force on the system. We model these forced systems with the nonhomogeneous differential equation

$$mx''+bx'+kx = f(t),$$

where the external force is represented by the $f(t)$ term. As we saw in Nonhomogenous Linear Equations, differential equations such as this have solutions of the form

$$x(t) = c_1x_1(t) + c_2x_2(t) + x_p(t),$$

where $c_1x_1(t) + c_2x_2(t)$ is the general solution to the complementary equation and $x_p(t)$ is a particular solution to the nonhomogeneous equation. If the system is damped, $\lim_{t\to\infty} c_1x_1(t) + c_2x_2(t) = 0$. Since these terms do not affect the long-term behavior of the system, we call this part of the solution the ***transient solution*.** The long-term behavior of the system is determined by $x_p(t)$, so we call this part of the solution the **steady-state solution**.

Example 17.3.7: Forced Vibrations

A mass of 1 slug stretches a spring 2 ft and comes to rest at equilibrium. The system is attached to a dashpot that imparts a damping force equal to eight times the instantaneous velocity of the mass. Find the equation of motion if an external force equal to $f(t) = 8\sin(4t)$ is applied to the system beginning at time $t = 0$. What is the transient solution? What is the steady-state solution?

Solution

We have $mg = 1(32) = 2k$, so $k = 16$ and the differential equation is

$$x''+8x'+16x = 8\sin(4t).$$

The general solution to the complementary equation is

$$c_1e^{-4t} + c_2te^{-4t}.$$

Assuming a particular solution of the form $x_p(t) = A\cos(4t) + B\sin(4t)$ and using the method of undetermined coefficients, we find $x_p(t) = -\frac{1}{4}\cos(4t)$, so

$$x(t) = c_1e^{-4t} + c_2te^{-4t} - \frac{1}{4}\cos(4t).$$

At $t = 0$, the mass is at rest in the equilibrium position, so $x(0) = x'(0) = 0$. Applying these initial conditions to solve for c_1 and c_2, we get

$$x(t) = \frac{1}{4}e^{-4t} + te^{-4t} - \frac{1}{4}\cos(4t).$$

The transient solution is $\frac{1}{4}e^{-4t} + te^{-4t}$. The steady-state solution is $-\frac{1}{4}\cos(4t)$.

Exercise 17.3.6

A mass of 2 kg is attached to a spring with constant 32 N/m and comes to rest in the equilibrium position. Beginning at time $t = 0$, an external force equal to $f(t) = 68e^{-2t}\cos(4t)$ is applied to the system. Find the equation of motion if there is no damping. What is the transient solution? What is the steady-state solution?

Hint

Find the particular solution *before* applying the initial conditions.

Answer

$x(t) = -\frac{1}{2}\cos(4t) + \frac{9}{4}\sin(4t) + \frac{1}{2}e^{-2t}\cos(4t) - 2e^{-2t}\sin(4t)$

Transient solution: $\frac{1}{2}e^{-2t}\cos(4t) - 2e^{-2t}\sin(4t)$

Steady-state solution: $-\frac{1}{2}\cos(4t) + \frac{9}{4}\sin(4t)$

RESONANCE

Consider an undamped system exhibiting simple harmonic motion. In the real world, we never truly have an undamped system; –some damping always occurs. For theoretical purposes, however, we could imagine a spring-mass system contained in a vacuum chamber. With no air resistance, the mass would continue to move up and down indefinitely.

The frequency of the resulting motion, given by $f = \frac{1}{T} = \frac{\omega}{2\pi}$, is called the *natural frequency of the system*. If an external force acting on the system has a frequency close to the natural frequency of the system, a phenomenon called ***resonance*** results. The external force reinforces and amplifies the natural motion of the system.

1. Consider the differential equation $x'' + x = 0$. Find the general solution. What is the natural frequency of the system?
2. Now suppose this system is subjected to an external force given by $f(t) = 5\cos t$. Solve the initial-value problem $x'' + x = 5\cos t$, $x(0) = 0$, $x'(0) = 1$.
3. Graph the solution. What happens to the behavior of the system over time?
4. In the real world, there is always some damping. However, if the damping force is weak, and the external force is strong enough, real-world systems can still exhibit resonance. One of the most famous examples of resonance is the collapse of the **Tacoma Narrows Bridge** on November 7, 1940. The bridge had exhibited strange behavior ever since it was built. The roadway had a strange "bounce" to it. On the day it collapsed, a strong windstorm caused the roadway to twist and ripple violently. The bridge was unable to withstand these forces and it ultimately collapsed. Experts believe the windstorm exerted forces on the bridge that were very close to its natural frequency, and the resulting resonance ultimately shook the bridge apart.

   This website contains more information about the collapse of the Tacoma Narrows Bridge.

   During the short time the Tacoma Narrows Bridge stood, it became quite a tourist attraction. Several people were on site the day the bridge collapsed, and one of them caught the collapse on film.



Watch the video to see the collapse of the Tacoma Narrows Bridge "Gallopin' Gertie". https://www.youtube.com/watch?v=j-zczJXSxnw

1. Another real-world example of resonance is a singer shattering a crystal wineglass when she sings just the right note. When someone taps a crystal wineglass or wets a finger and runs it around the rim, a tone can be heard. That note is created by the wineglass vibrating at its natural frequency. If a singer then sings that same note at a high enough volume, the glass shatters as a result of resonance.

   The TV show *Mythbusters* aired an episode on this phenomenon. Visit this website to learn more about it. Adam Savage also described the experience. Watch this video for his account.

The *RLC* Series Circuit

Consider an electrical circuit containing a resistor, an inductor, and a capacitor, as shown in Figure 17.3.12. Such a circuit is called an ***RLC* series circuit**. *RLC* circuits are used in many electronic systems, most notably as tuners in AM/FM radios. The tuning knob varies the capacitance of the capacitor, which in turn tunes the radio. Such circuits can be modeled by second-order, constant-coefficient differential equations.

Let $I(t)$ denote the current in the *RLC* circuit and $q(t)$ denote the charge on the capacitor. Furthermore, let L denote inductance in henrys (H), R denote resistance in ohms (Ω), and C denote capacitance in farads (F). Last, let $E(t)$ denote electric potential in volts (V).

Kirchhoff's voltage rule states that the sum of the voltage drops around any closed loop must be zero. So, we need to consider the voltage drops across the inductor (denoted E_L), the resistor (denoted E_R), and the capacitor (denoted E_C). Because the *RLC* circuit shown in Figure 17.3.12 includes a voltage source, $E(t)$, which adds voltage to the circuit, we have $E_L + E_R + E_C = E(t)$.

We present the formulas below without further development and those of you interested in the derivation of these formulas can review the links. Using Faraday's law and Lenz's law, the voltage drop across an inductor can be shown to be proportional to the instantaneous rate of change of current, with proportionality constant L. Thus,

$$E_L = L\frac{dI}{dt}.$$

Next, according to Ohm's law, the voltage drop across a resistor is proportional to the current passing through the resistor, with proportionality constant R. Therefore,

$$E_R = RI.$$

Last, the voltage drop across a capacitor is proportional to the charge, q, on the capacitor, with proportionality constant $1/C$. Thus,

$$E_C = \frac{1}{C}q.$$

Adding these terms together, we get

$$L\frac{dI}{dt} + RI + \frac{1}{C}q = E(t).$$

Noting that $I = (dq)/(dt)$, this becomes

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t).$$

Mathematically, this system is analogous to the spring-mass systems we have been examining in this section.

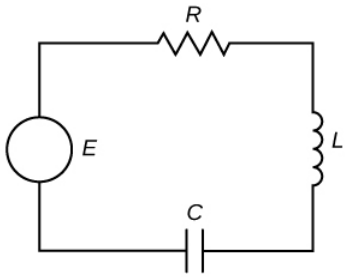


Figure 17.3.12: An *RLC* series circuit can be modeled by the same differential equation as a mass-spring system.

Series Circuit

Find the charge on the capacitor in an *RLC* series circuit where $L = 5/3$ H, $R = 10\Omega$, $C = 1/30$ F, and $E(t) = 300$ V. Assume the initial charge on the capacitor is 0 C and the initial current is 9 A. What happens to the charge on the capacitor over time?

Solution

We have

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$
$$\frac{5}{3}\frac{d^2q}{dt^2} + 10\frac{dq}{dt} + 30q = 300$$
$$\frac{d^2q}{dt^2} + 6\frac{dq}{dt} + 18q = 180.$$

The general solution to the complementary equation is

$$e^{-3t}(c_1\cos(3t) + c_2\sin(3t)).$$

Assume a particular solution of the form $q_p = A$, where A is a constant. Using the method of undetermined coefficients, we find $A = 10$. So,

$$q(t) = e^{-3t}(c_1\cos(3t) + c_2\sin(3t)) + 10.$$

Applying the initial conditions $q(0) = 0$ and $i(0) = ((dq)/(dt))(0) = 9$, we find $c_1 = -10$ and $c_2 = -7$. So the charge on the capacitor is

$$q(t) = -10e^{-3t}\cos(3t) - 7e^{-3t}\sin(3t) + 10.$$

Looking closely at this function, we see the first two terms will decay over time (as a result of the negative exponent in the exponential function). Therefore, the capacitor eventually approaches a steady-state charge of 10 C.

Exercise 17.3.7

Find the charge on the capacitor in an *RLC* series circuit where $L = 1/5$ H, $R = 2/5\Omega$, $C = 1/2$ F, and $E(t) = 50$ V. Assume the initial charge on the capacitor is 0 C and the initial current is 4 A.

Hint

Remember, $E_L = L((dI)/(dt))$.

Answer

$$q(t) = -25e^{-t}\cos(3t) - 7e^{-t}\sin(3t) + 25$$

Key Concepts

- Second-order constant-coefficient differential equations can be used to model spring-mass systems.
- An examination of the forces on a spring-mass system results in a differential equation of the form

  $$mx''+bx'+kx = f(t),$$

  where m represents the mass, b is the coefficient of the damping force, k is the spring constant, and $f(t)$ represents any net external forces on the system.
- If $b = 0$, there is no damping force acting on the system, and simple harmonic motion results.
- If $b \neq 0$,the behavior of the system depends on whether $b^2 - 4mk > 0, b^2 - 4mk = 0$, or $b^2 - 4mk < 0$.
  - If $b^2 - 4mk > 0$, the system is overdamped and does not exhibit oscillatory behavior.
  - If $b^2 - 4mk = 0$, the system is critically damped. It does not exhibit oscillatory behavior, but any slight reduction in the damping would result in oscillatory behavior.
  - If $b^2 - 4mk < 0$, the system is underdamped. It exhibits oscillatory behavior, but the amplitude of the oscillations decreases over time.
- If$f(t) \neq 0$, the solution to the differential equation is the sum of a transient solution and a steady-state solution. The steady-state solution governs the long-term behavior of the system.
- The charge on the capacitor in an *RLC* series circuit can also be modeled with a second-order constant-coefficient differential equation of the form

  $$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t),$$

  where L is the inductance, R is the resistance, C is the capacitance, and $E(t)$ is the voltage source.

Key Equations

- Equation of simple harmonic motion

  $$x''+\omega^2 x = 0$$

- Solution for simple harmonic motion

  $$x(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$$

- Alternative form of solution for SHM

  $$x(t) = A\sin(\omega t + \phi)$$

- Forced harmonic motion

  $$mx''+bx'+kx = f(t)$$

- Charge in a RLC series circuit

  $$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t),$$

Glossary

RLC series circuit

a complete electrical path consisting of a resistor, an inductor, and a capacitor; a second-order, constant-coefficient differential equation can be used to model the charge on the capacitor in an *RLC* series circuit

simple harmonic motion

motion described by the equation $x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$, as exhibited by an undamped spring-mass system in which the mass continues to oscillate indefinitely

steady-state solution

a solution to a nonhomogeneous differential equation related to the forcing function; in the long term, the solution approaches the steady-state solution

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17.3E: Exercises for Section 17.3

1. A mass weighing 4 lb stretches a spring 8 in. Find the equation of motion if the spring is released from the equilibrium position with a downward velocity of 12 ft/sec. What is the period and frequency of the motion?

2. A mass weighing 2 lb stretches a spring 2 ft. Find the equation of motion if the spring is released from 2 in. below the equilibrium position with an upward velocity of 8 ft/sec. What is the period and frequency of the motion?

Answer

differential equation: $x''+16x=0$,
equation of motion: $x(t)=\frac{1}{6}\cos(4t)-2\sin(4t)$,
period $=\frac{\pi}{2}$ sec,
frequency $=\frac{2}{\pi}$ Hz

3. A 100-g mass stretches a spring 0.1 m. Find the equation of motion of the mass if it is released from rest from a position 20 cm below the equilibrium position. What is the frequency of this motion?

4. A 400-g mass stretches a spring 5 cm. Find the equation of motion of the mass if it is released from rest from a position 15 cm below the equilibrium position. What is the frequency of this motion?

Answer

differential equation: $x''+196x=0$,
equation of motion: $x(t)=0.15\cos(14t)$,
period $=\frac{\pi}{7}$ sec,
frequency $=\frac{7}{\pi}$ Hz

5. A block has a mass of 9 kg and is attached to a vertical spring with a spring constant of 0.25 N/m. The block is stretched 0.75 m below its equilibrium position and released.

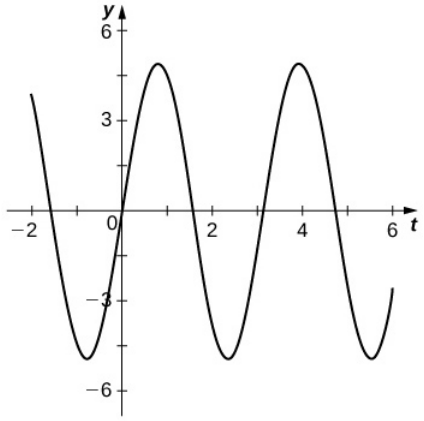
a. Find the position function $x(t)$ of the block.
b. Find the period and frequency of the vibration.
c. Sketch a graph of $x(t)$.
d. At what time does the block first pass through the equilibrium position?

6. A block has a mass of 5 kg and is attached to a vertical spring with a spring constant of 20 N/m. The block is released from the equilibrium position with a downward velocity of 10 m/sec.

a. Find the position function $x(t)$ of the block.
b. Find the period and frequency of the vibration.
c. Sketch a graph of $x(t)$.
d. At what time does the block first pass through the equilibrium position?

Answer

a. $x(t)=5\sin(2t)$
b. period $=\pi$ sec,
frequency $=\frac{1}{\pi}$ Hz
c.



d. $t = \frac{\pi}{2}$ sec

7. A 1-kg mass is attached to a vertical spring with a spring constant of 21 N/m. The resistance in the spring-mass system is equal to 10 times the instantaneous velocity of the mass.

a. Find the equation of motion if the mass is released from a position 2 m below its equilibrium position with a downward velocity of 2 m/sec.
b. Graph the solution and determine whether the motion is overdamped, critically damped, or underdamped.

8. An 800-lb weight (25 slugs) is attached to a vertical spring with a spring constant of 226 lb/ft. The system is immersed in a medium that imparts a damping force equal to 10 times the instantaneous velocity of the mass.

a. Find the equation of motion if it is released from a position 20 ft below its equilibrium position with a downward velocity of 41 ft/sec.
b. Graph the solution and determine whether the motion is overdamped, critically damped, or underdamped.

Answer

a. $x(t) = e^{-t/5}(20\cos(3t) + 15\sin(3t))$

b. underdamped

9. A 9-kg mass is attached to a vertical spring with a spring constant of 16 N/m. The system is immersed in a medium that imparts a damping force equal to 24 times the instantaneous velocity of the mass.

a. Find the equation of motion if it is released from its equilibrium position with an upward velocity of 4 m/sec.
b. Graph the solution and determine whether the motion is overdamped, critically damped, or underdamped.

10. A 1-kg mass stretches a spring 6.25 cm. The resistance in the spring-mass system is equal to eight times the instantaneous velocity of the mass.

a. Find the equation of motion if the mass is released from a position 5 m below its equilibrium position with an upward velocity of 10 m/sec.
b. Determine whether the motion is overdamped, critically damped, or underdamped.

Answer

a. $x(t) = 5e^{-4t} + 10te^{-4t}$

b. critically damped

11. A 32-lb weight (1 slug) stretches a vertical spring 4 in. The resistance in the spring-mass system is equal to four times the instantaneous velocity of the mass.

a. Find the equation of motion if it is released from its equilibrium position with a downward velocity of 12 ft/sec.
b. Determine whether the motion is overdamped, critically damped, or underdamped.

12. A 64-lb weight is attached to a vertical spring with a spring constant of 4.625 lb/ft. The resistance in the spring-mass system is equal to the instantaneous velocity. The weight is set in motion from a position 1 ft below its equilibrium position with an upward velocity of 2 ft/sec. Is the mass above or below the equation position at the end of π sec? By what distance?

Answer

$x(\pi) = \frac{7e^{-\pi/4}}{6}$ ft below

13. A mass that weighs 8 lb stretches a spring 6 inches. The system is acted on by an external force of $8\sin 8t$lb. If the mass is pulled down 3 inches and then released, determine the position of the mass at any time.

14. A mass that weighs 6 lb stretches a spring 3 in. The system is acted on by an external force of $8\sin(4t)$ lb. If the mass is pulled down 1 inch and then released, determine the position of the mass at any time.

Answer

$x(t) = \frac{32}{9}\sin(4t) + \cos(\sqrt{128}t) - \frac{16}{9\sqrt{2}}\sin(\sqrt{128}t)$

15. Find the charge on the capacitor in an *RLC* series circuit where $L = 40$ H, $R = 30\,\Omega$, $C = 1/200$ F, and $E(t) = 200$ V. Assume the initial charge on the capacitor is 7 C and the initial current is 0 A.

16. Find the charge on the capacitor in an *RLC* series circuit where $L = 2$ H, $R = 24\,\Omega$, $C = 0.005$ F, and $E(t) = 12\sin 10t$ V. Assume the initial charge on the capacitor is 0.001 C and the initial current is 0 A.

Answer

$q(t) = e^{-6t}(0.051\cos(8t) + 0.03825\sin(8t)) - \frac{1}{20}\cos(10t)$

17. A series circuit consists of a device where$L = 1$ H, $R = 20\,\Omega$, $C = 0.002$ F, and $E(t) = 12$ V. If the initial charge and current are both zero, find the charge and current at time t.

18. A series circuit consists of a device where $L = 12$ H, $R = 10\,\Omega$, $C = \frac{1}{50}$ F, and $E(t) = 250$ V. If the initial charge on the capacitor is 0 C and the initial current is 18 A, find the charge and current at time t.

Answer

$q(t) = e^{-10t}(-32t - 5) + 5, I(t) = 2e^{-10t}(160t + 9)$

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17.4: Series Solutions of Differential Equations

Learning Objectives

- Use power series to solve first-order and second-order differential equations.

Previously, we studied how functions can be represented as power series, $y(x) = \sum_{n=0}^{\infty} a_n x^n$. We also saw that we can find series representations of the derivatives of such functions by differentiating the power series term by term. This gives

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

and

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

In some cases, these power series representations can be used to find solutions to differential equations.

The examples and exercises in this section were chosen for which power solutions exist. However, it is not always the case that power solutions exist. Those of you interested in a more rigorous treatment of this topic should review the differential equations section of the LibreTexts.

Problem-Solving Strategy: Finding Power Series Solutions to Differential Equations

1. Assume the differential equation has a solution of the form

$$y(x) = \sum_{n=0}^{\infty} a_n x^n.$$

2. Differentiate the power series term by term to get

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

and

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

3. Substitute the power series expressions into the differential equation.
4. Re-index sums as necessary to combine terms and simplify the expression.
5. Equate coefficients of like powers of x to determine values for the coefficients a_n in the power series.
6. Substitute the coefficients back into the power series and write the solution.

Example 17.4.1: Series Solutions to Differential Equations

Find a power series solution for the following differential equations.

a. $y'' - y = 0$
b. $(x^2 - 1)y'' + 6xy' + 4y = -4$

Solution

Part a

Assume

$$y(x) = \sum_{n=0}^{\infty} a_n x^n \qquad \text{(step 1)}$$

Then,

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \qquad \text{(step 2A)}$$

and

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \qquad \text{(step 2B)}$$

We want to find values for the coefficients a_n such that

$$y'' - y = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^n = 0. \qquad \text{(step 3)}$$

We want the indices on our sums to match so that we can express them using a single summation. That is, we want to rewrite the first summation so that it starts with $n = 0$.

To re-index the first term, replace n with $n + 2$ inside the sum, and change the lower summation limit to $n = 0$. We get

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

This gives

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=0}^{\infty} a_n x_n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} - a_n] x^n = 0. \qquad \text{(step 4)}$$

Because power series expansions of functions are unique, this equation can be true only if the coefficients of each power of x are zero. So we have

$$(n+2)(n+1) a_{n+2} - a_n = 0 \text{ for } n = 0, 1, 2, \ldots .$$

This recurrence relationship allows us to express each coefficient a_n in terms of the coefficient two terms earlier. This yields one expression for even values of n and another expression for odd values of n. Looking first at the equations involving even values of n, we see that

$$a_2 = \frac{a_0}{2}$$

$$a_4 = \frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}$$

$$a_6 = \frac{a_4}{6 \cdot 5} = \frac{a_0}{6!}$$

$$\vdots$$

Thus, in general, when n is even,

$$a_n = \frac{a_0}{n!}. \qquad \text{(step 5)}$$

For the equations involving odd values of n, we see that

$$a_3 = \frac{a_1}{3 \cdot 2} = \frac{a_1}{3!}$$
$$a_5 = \frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}$$
$$a_7 = \frac{a_5}{7 \cdot 6} = \frac{a_1}{7!}$$
$$\vdots$$

Therefore, in general, when n is odd,

$$a_n = \frac{a_1}{n!}. \qquad \text{(step 5)}$$

Putting this together, we have

$$\begin{aligned} y(x) &= \sum_{n=0}^{\infty} a_n x^n \\ &= a_0 + a_1 x + \frac{a_0}{2} x^2 + \frac{a_1}{3!} x^3 + \frac{a_0}{4!} x^4 + \frac{a_1}{5!} x^5 + \cdots. \end{aligned}$$

Re-indexing the sums to account for the even and odd values of n separately, we obtain

$$y(x) = a_0 \sum_{k=0}^{\infty} \frac{1}{(2k)!} x^{2k} + a_1 \sum_{k=0}^{\infty} \frac{1}{(2k+1)!} x^{2k+1}. \qquad \text{(step 6)}$$

Analysis for part a.

As expected for a second-order differential equation, this solution depends on two arbitrary constants. However, note that our differential equation is a constant-coefficient differential equation, yet the power series solution does not appear to have the familiar form (containing exponential functions) that we are used to seeing. Furthermore, since $y(x) = c_1 e^x + c_2 e^{-x}$ is the general solution to this equation, we must be able to write any solution in this form, and it is not clear whether the power series solution we just found can, in fact, be written in that form.

Fortunately, after writing the power series representations of e^x and e^{-x}, and doing some algebra, we find that if we choose

$$c_0 = \frac{(a_0 + a_1)}{2}, c_1 = \frac{(a_0 - a_1)}{2},$$

we then have $a_0 = c_0 + c_1$ and $a_1 = c_0 - c_1$, and

$$\begin{aligned} y(x) &= a_0 + a_1 x + \frac{a_0}{2} x^2 + \frac{a_1}{3!} x^3 + \frac{a_0}{4!} x^4 + \frac{a_1}{5!} x^5 + \cdots \\ &= (c_0 + c_1) + (c_0 - c_1) x + \frac{(c_0 + c_1)}{2} x^2 + \frac{(c_0 - c_1)}{3!} x^3 + \frac{(c_0 + c_1)}{4!} x^4 + \frac{(c_0 - c_1)}{5!} x^5 + \cdots \\ &= c_0 \sum_{n=0}^{\infty} \frac{x^n}{n!} + c_1 \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} \\ &= c_0 e^x + c_1 e^{-x}. \end{aligned}$$

So we have, in fact, found the same general solution. Note that this choice of c_1 and c_2 is not obvious. This is a case when we know what the answer should be, and have essentially "reverse-engineered" our choice of coefficients.

Part b

Assume

$$y(x) = \sum_{n=0}^{\infty} a_n x^n \qquad \text{(step 1)}$$

Then,

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \qquad \text{(step 2)}$$

and

$$y''(x) = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} \qquad \text{(step 2)}$$

We want to find values for the coefficients a_n such that

$$\begin{aligned}(x^2-1)y''+6xy'+4y &= -4\\ (x^2-1)\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 6x\sum_{n=1}^{\infty} na_n x^{n-1} + 4\sum_{n=0}^{\infty} a_n x^n &= -4\\ x^2\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 6x\sum_{n=1}^{\infty} na_n x^{n-1} + 4\sum_{n=0}^{\infty} a_n x^n &= -4.\end{aligned}$$

Taking the external factors inside the summations, we get

$$\sum_{n=2}^{\infty} n(n-1)a_n x^n - \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=1}^{\infty} 6na_n x^n + \sum_{n=0}^{\infty} 4a_n x^n = -4. \qquad \text{(step 3)}$$

Now, in the first summation, we see that when $n = 0$ or $n = 1$, the term evaluates to zero, so we can add these terms back into our sum to get

$$\sum_{n=2}^{\infty} n(n-1)a_n x^n = \sum_{n=0}^{\infty} n(n-1)a_n x^n.$$

Similarly, in the third term, we see that when $n = 0$, the expression evaluates to zero, so we can add that term back in as well. We have

$$\sum_{n=1}^{\infty} 6na_n x^n = \sum_{n=0}^{\infty} 6na_n x^n.$$

Then, we need only shift the indices in our second term. We get

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n.$$

Thus, we have

$$\begin{aligned}\sum_{n=0}^{\infty} n(n-1)a_n x^n - \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + \sum_{n=0}^{\infty} 6na_n x^n + \sum_{n=0}^{\infty} 4a_n x^n &= -4 \qquad \text{(step 4)}\\ \sum_{n=0}^{\infty} [n(n-1)a_n - (n+2)(n+1)a_{n+2} + 6na_n + 4a_n]x^n &= -4\\ \sum_{n=0}^{\infty} [(n^2-n)a_n + 6na_n + 4a_n - (n+2)(n+1)a_{n+2}]x^n &= -4\\ \sum_{n=0}^{\infty} [n^2 a_n + 5na_n + 4a_n - (n+2)(n+1)a_{n+2}]x^n &= -4\\ \sum_{n=0}^{\infty} [(n^2+5n+4)a_n - (n+2)(n+1)a_{n+2}]x^n &= -4\\ \sum_{n=0}^{\infty} [(n+4)(n+1)a_n - (n+2)(n+1)a_{n+2}]x^n &= -4\end{aligned}$$

Looking at the coefficients of each power of x, we see that the constant term must be equal to -4, and the coefficients of all other powers of x must be zero. Then, looking first at the constant term,

$$\begin{aligned}4a_0 - 2a_2 &= -4\\ a_2 &= 2a_0 + 2\end{aligned} \qquad \text{(step 3)}$$

For $n \geq 1$, we have

$$(n+4)(n+1)a_n-(n+2)(n+1)a_{n+2}=0$$
$$(n+1)[(n+4)a_n-(n+2)a_{n+2}]=0.$$

Since $n\geq 1,\ n+1\neq 0$, we see that

$$(n+4)a_n-(n+2)a_{n+2}=0$$

and thus

$$a_{n+2}=\frac{n+4}{n+2}a_n.$$

For even values of n, we have

$$a_4=\frac{6}{4}(2a_0+2)=3a_0+3$$
$$a_6=\frac{8}{6}(3a_0+3)=4a_0+4$$
$$\vdots$$

In general,

$$a_{2k}=(k+1)(a_0+1). \qquad \text{(step 5)}$$

For odd values of n, we have

$$a_3=\frac{5}{3}a_1$$
$$a_5=\frac{7}{5}a_3=\frac{7}{3}a_1$$
$$a_7=\frac{9}{7}a_5=\frac{9}{3}a_1=3a_1$$
$$\vdots$$

In general,

$$a_{2k+1}=\frac{2k+3}{3}a_1. \qquad \text{(step 5 continued)}$$

Putting this together, we have

$$y(x)=\sum_{k=0}^{\infty}(k+1)(a_0+1)x^{2k}+\sum_{k=0}^{\infty}(\frac{2k+3}{3})a_1x^{2k+1}. \qquad \text{(step 6)}$$

Exercise 17.4.1

Find a power series solution for the following differential equations.

a. $y'+2xy=0$
b. $(x+1)y'=3y$

Hint

Follow the problem-solving strategy.

Answer a

$$y(x)=a_0\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}x^{2n}=a_0e^{-x^2}$$

Answer b

$$y(x)=a_0(x+1)^3$$

Bessel functions

We close this section with a brief introduction to *Bessel functions*. Complete treatment of Bessel functions is well beyond the scope of this course, but we get a little taste of the topic here so we can see how series solutions to differential equations are used in real-world applications. The Bessel equation of order n is given by

$$x^2y''+xy'+(x^2-n^2)y=0.$$

This equation arises in many physical applications, particularly those involving cylindrical coordinates, such as the vibration of a circular drum head and transient heating or cooling of a cylinder. In the next example, we find a power series solution to the Bessel equation of order 0.

Example 17.4.2: Power Series Solution to the Bessel Equation

Find a power series solution to the Bessel equation of order 0 and graph the solution.

Solution

The Bessel equation of order 0 is given by

$$x^2y''+xy'+x^2y=0.$$

We assume a solution of the form $y=\sum_{n=0}^{\infty}a_nx^n$. Then $y'(x)=\sum_{n=1}^{\infty}na_nx^{n-1}$ and $y''(x)=\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}$.Substituting this into the differential equation, we get

$$\begin{aligned}
x^2\sum_{n=2}^{\infty}n(n-1)a_nx^{n-2}+x\sum_{n=1}^{\infty}na_nx^{n-1}&\\
+x^2\sum_{n=0}^{\infty}a_nx^n=0 &\quad \text{Substitution.}\\
\sum_{n=2}^{\infty}n(n-1)a_nx^n+\sum_{n=1}^{\infty}na_nx^n+\sum_{n=0}^{\infty}a_nx^{n+2}=0 &\quad \text{Bring external factors within sums.}\\
\sum_{n=2}^{\infty}n(n-1)a_nx^n+\sum_{n=1}^{\infty}na_nx^n+\sum_{n=2}^{\infty}a_{n-2}x^n=0 &\quad \text{Re-index third sum.}\\
\sum_{n=2}^{\infty}n(n-1)a_nx^n+a_1x&\\
+\sum_{n=2}^{\infty}na_nx^n+\sum_{n=2}^{\infty}a_{n-2}x^n=0 &\quad \text{Separate } n=1 \text{ term from second sum.}\\
a_1x+\sum_{n=2}^{\infty}[n(n-1)a_n+na_n+a_{n-2}]x^n=0 &\quad \text{Collect summation terms.}\\
a_1x+\sum_{n=2}^{\infty}[(n^2-n)a_n+na_n+a_{n-2}]x^n=0 &\quad \text{Multiply through in first term.}\\
a_1x+\sum_{n=2}^{\infty}[n^2a_n+a_{n-2}]x^n=0. &\quad \text{Simplify.}
\end{aligned}$$

Then, $a_1=0$, and for $n\geq 2$,

$$\begin{aligned}
n^2a_n+a_{n-2}&=0\\
a_n&=-\frac{1}{n^2}a_{n-2}.
\end{aligned}$$

Because $a_1=0$, all odd terms are zero. Then, for even values of n, we have

$$a_2 = -\frac{1}{2^2}a_0$$
$$a_4 = -\frac{1}{4^2}a_2 = \frac{1}{4^2 \cdot 2^2}a_0.$$
$$a_6 = -\frac{1}{6^2}a_4 = -\frac{1}{6^2 \cdot 4^2 \cdot 2^2}a_0$$

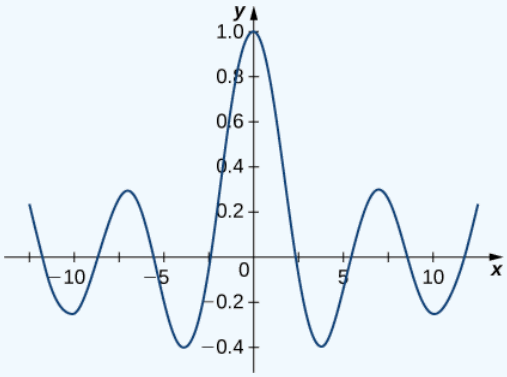
In general,

$$a_{2k} = \frac{(-1)^k}{(2)^{2k}(k!)^2}a_0.$$

Thus, we have

$$y(x) = a_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2)^{2k}(k!)^2}x^{2k}.$$

The graph appears below.



Exercise 17.4.2

Verify that the expression found in Example 17.4.2 is a solution to the Bessel equation of order 0.

Hint

Differentiate the power series term by term and substitute it into the differential equation.

Key Concepts

- Power series representations of functions can sometimes be used to find solutions to differential equations.
- Differentiate the power series term by term and substitute into the differential equation to find relationships between the power series coefficients.

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17.4E: Exercises for Section 17.4

Find a power series solution for each of the following differential equations.

1. $y''+6y'=0$

2. $5y''+y'=0$

Answer

$$y=c_0+5c_1\sum_{n=1}^{\infty}\frac{(-x/5)^n}{n!}=c_0+5c_1e^{-x/5}$$

3. $y''+25y=0$

4. $y''-y=0$

Answer

$$y=c_0\sum_{n=0}^{\infty}\frac{(x)^{2n}}{(2n)!}+c_1\sum_{n=0}^{\infty}\frac{(x)^{2n+1}}{(2n+1)!}$$

5. $2y'+y=0$

6. $y'-2xy=0$

Answer

$$y=c_0\sum_{n=0}^{\infty}\frac{x^{2n}}{n!}=c_0e^{x^2}$$

7. $(x-7)y'+2y=0$

8. $y''-xy'-y=0$

Answer

$$y=c_0\sum_{n=0}^{\infty}\frac{x^{2n}}{2^n n!}+c_1\sum_{n=0}^{\infty}\frac{x^{2n+1}}{1\cdot 3\cdot 5\cdot 7\cdots(2n+1)}$$

9. $(1+x^2)y''-4xy'+6y=0$

10. $x^2y''-xy'-3y=0$

Answer

$$y=c_1x^3+\frac{c_2}{x}$$

11. $y''-8y'=0,\quad y(0)=-2,\ y'(0)=10$

12. $y''-2xy=0,\quad y(0)=1,\ y'(0)=-3$

Answer

$$y=1-3x+\frac{2x^3}{3!}-\frac{12x^4}{4!}+\frac{16x^6}{6!}-\frac{120x^7}{7!}+\cdots$$

13. The differential equation $x^2y''+xy'+(x^2-1)y=0$ is a Bessel equation of order 1. Use a power series of the form $y=\sum_{n=0}^{\infty}a_nx^n$ to find the solution.

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17.R: Chapter 17 Review Exercises

***True or False*? Justify your answer with a proof or a counterexample.**

1. If y and z are both solutions to $y''+2y'+y=0$, then $y+z$ is also a solution.

Answer

True

2. The following system of algebraic equations has a unique solution:

$6z_1+3z_2=8$
$4z_1+2z_2=4.$

3. $y=e^x\cos(3x)+e^x\sin(2x)$ is a solution to the second-order differential equation $y''+2y'+10=0.$

Answer

False

4. To find the particular solution to a second-order differential equation, you need one initial condition.

In problems 5 - 8, classify the differential equations. Determine the order, whether it is linear and, if linear, whether the differential equation is homogeneous or nonhomogeneous. If the equation is second-order homogeneous and linear, find the characteristic equation.

5. $y''-2y=0$

Answer

second order, linear, homogeneous, $\lambda^2-2=0$

6. $y''-3y+2y=\cos(t)$

7. $\left(\frac{dy}{dt}\right)^2+yy'=1$

Answer

first order, nonlinear, nonhomogeneous

8. $\frac{d^2y}{dt^2}+t\frac{dy}{dt}+\sin^2(t)y=e^t$

In problems 9 - 16, find the general solution.

9. $y''+9y=0$

Answer

$y=c_1\sin(3x)+c_2\cos(3x)$

10. $y''+2y'+y=0$

11. $y''-2y'+10y=4x$

Answer

$y=c_1e^x\sin(3x)+c_2e^x\cos(3x)+\frac{2}{5}x+\frac{2}{25}$

12. $y''=\cos(x)+2y'+y$

13. $y''+5y+y=x+e^{2x}$

Answer

$y=c_1e^{-x}+c_2e^{-4x}+\frac{x}{4}+\frac{e^{2x}}{18}-\frac{5}{16}$

14. $y'' = 3y' + xe^{-x}$

15. $y'' - x^2 = -3y' - \frac{9}{4}y + 3x$

Answer

$y = c_1 e^{(-3/2)x} + c_2 x e^{(-3/2)x} + \frac{4}{9}x^2 + \frac{4}{27}x - \frac{16}{27}$

16. $y'' = 2\cos x + y' - y$

In problems 17 - 18, find the solution to the initial-value problem, if possible.

17. $y'' + 4y' + 6y = 0,\ y(0) = 0,\ y'(0) = \sqrt{2}$

Answer

$y = e^{-2x}\sin(\sqrt{2}x)$

18. $y'' = 3y - \cos(x),\ y(0) = \frac{9}{4},\ y'(0) = 0$

In problems 19 - 20, find the solution to the boundary-value problem.

19. $4y' = -6y + 2y'',\ y(0) = 0,\ y(1) = 1$

Answer

$y = \dfrac{e^{1-x}}{e^4 - 1}(e^{4x} - 1)$

20. $y'' = 3x - y - y',\ y(0) = -3,\ y(1) = 0$

For the following problem, set up and solve the differential equation.

21. The motion of a swinging pendulum for small angles θ can be approximated by $\dfrac{d^2\theta}{dt^2} + \dfrac{g}{L}\theta = 0$, where θ is the angle the pendulum makes with respect to a vertical line, g is the acceleration resulting from gravity, and L is the length of the pendulum. Find the equation describing the angle of the pendulum at time t, assuming an initial displacement of θ_0 and an initial velocity of zero.

Answer

$\theta(t) = \theta_0 \cos\left(\sqrt{\frac{g}{l}}t\right)$

In problems 22 - 23, consider the "beats" that occur when the forcing term of a differential equation causes "slow" and "fast" amplitudes. Consider the general differential equation $ay'' + by = \cos(\omega t)$ that governs undamped motion. Assume that $\sqrt{\frac{b}{a}} \neq \omega$.

22. Find the general solution to this equation (*Hint:* call $\omega_0 = \sqrt{b/a}$).

23. Assuming the system starts from rest, show that the particular solution can be written as $y = \dfrac{2}{a(\omega_0^2 - \omega^2)}\sin\left(\dfrac{\omega_0 - \omega t}{2}\right)\sin\left(\dfrac{\omega_0 + \omega t}{2}\right)$.

24. [T] Using your solutions derived earlier, plot the solution to the system $2y'' + 9y = \cos(2t)$ over the interval $t = [-50, 50]$. Find, analytically, the period of the fast and slow amplitudes.

For the following problem, set up and solve the differential equations.

25. An opera singer is attempting to shatter a glass by singing a particular note. The vibrations of the glass can be modeled by $y'' + ay = \cos(bt)$, where $y'' + ay = 0$ represents the natural frequency of the glass and the singer is forcing the vibrations at $\cos(bt)$. For what value b would the singer be able to break that glass? (*Note*: in order for the glass to break, the oscillations would need to get higher and higher.)

Answer

$b = \sqrt{a}$

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CHAPTER OVERVIEW

Appendices

Appendix A: Table of Derivatives
Appendix B: Table of Integrals

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Appendix A: Table of Derivatives

General Formulas

1. $\frac{d}{dx}(c) = 0$
2. $\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}(x^n) = nx^{n-1}, \quad \text{for real numbers } n$
5. $\frac{d}{dx}(cf(x)) = cf'(x)$
6. $\frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x)$
7. $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$
8. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Trigonometric Functions

9. $\frac{d}{dx}(\sin x) = \cos x$
10. $\frac{d}{dx}(\tan x) = \sec^2 x$
11. $\frac{d}{dx}(\sec x) = \sec x \tan x$
12. $\frac{d}{dx}(\cos x) = -\sin x$
13. $\frac{d}{dx}(\cot x) = -\csc^2 x$
14. $\frac{d}{dx}(\csc x) = -\csc x \cot x$

Inverse Trigonometric Functions

15. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
16. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
17. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
18. $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$
19. $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$
20. $\frac{d}{dx}(\csc^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$

Exponential and Logarithmic Functions

21. $\frac{d}{dx}(e^x) = e^x$
22. $\frac{d}{dx}(\ln|x|) = \frac{1}{x}$
23. $\frac{d}{dx}(b^x) = b^x \ln b$
24. $\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$

Hyperbolic Functions

25. $\frac{d}{dx}(\sinh x) = \cosh x$
26. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$
27. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
28. $\frac{d}{dx}(\cosh x) = \sinh x$
29. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$
30. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$

Inverse Hyperbolic Functions

31. $\frac{d}{dx}\left(\sinh^{-1} x\right) = \frac{1}{\sqrt{x^2+1}}$
32. $\frac{d}{dx}\left(\tanh^{-1} x\right) = \frac{1}{1-x^2} \quad (|x| < 1)$
33. $\frac{d}{dx}\left(\operatorname{sech}^{-1} x\right) = \frac{-1}{x\sqrt{1-x^2}} \quad (0 < x < 1)$
34. $\frac{d}{dx}\left(\cosh^{-1} x\right) = \frac{1}{\sqrt{x^2-1}} \quad (x > 1)$
35. $\frac{d}{dx}\left(\coth^{-1} x\right) = \frac{1}{1-x^2} \quad (|x| > 1)$
36. $\frac{d}{dx}\left(\operatorname{csch}^{-1} x\right) = \frac{-1}{|x|\sqrt{1+x^2}} \quad (x \neq 0)$

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Appendix B: Table of Integrals

Basic Integrals

1. $\int u^n\,du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1$
2. $\int \frac{du}{u} = \ln|u| + C$
3. $\int e^u\,du = e^u + C$
4. $\int a^u\,du = \frac{a^u}{\ln a} + C$
5. $\int \sin u\,du = -\cos u + C$
6. $\int \cos u\,du = \sin u + C$
7. $\int \sec^2 u\,du = \tan u + C$
8. $\int \csc^2 u\,du = -\cot u + C$
9. $\int \sec u \tan u\,du = \sec u + C$
10. $\int \csc u \cot u\,du = -\csc u + C$
11. $\int \tan u\,du = \ln|\sec u| + C$
12. $\int \cot u\,du = \ln|\sin u| + C$
13. $\int \sec u\,du = \ln|\sec u + \tan u| + C$
14. $\int \csc u\,du = \ln|\csc u - \cot u| + C$
15. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$
16. $\int \frac{du}{a^2 + u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$
17. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a}\sec^{-1}\frac{|u|}{a} + C$

Trigonometric Integrals

18. $\int \sin^2 u\,du = \frac{1}{2}u - \frac{1}{4}\sin 2u + C$
19. $\int \cos^2 u\,du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$
20. $\int \tan^2 u\,du = \tan u - u + C$
21. $\int \cot^2 u\,du = -\cot u - u + C$
22. $\int \sin^3 u\,du = -\frac{1}{3}(2 + \sin^2 u)\cos u + C$
23. $\int \cos^3 u\,du = \frac{1}{3}(2 + \cos^2 u)\sin u + C$
24. $\int \tan^3 u\,du = \frac{1}{2}\tan^2 u + \ln|\cos u| + C$
25. $\int \cot^3 u\,du = -\frac{1}{2}\cot^2 u - \ln|\sin u| + C$

26. $\int \sec^3 u\,du = \frac{1}{2}\sec u \tan u + \frac{1}{2}\ln|\sec u + \tan u| + C$

27. $\int \csc^3 u\,du = -\frac{1}{2}\csc u \cot u + \frac{1}{2}\ln|\csc u - \cot u| + C$

28. $\int \sin^n u\,du = \frac{-1}{n}\sin^{n-1} u \cos u + \frac{n-1}{n}\int \sin^{n-2} u\,du$

29. $\int \cos^n u\,du = \frac{1}{n}\cos^{n-1} u \sin u + \frac{n-1}{n}\int \cos^{n-2} u\,du$

30. $\int \tan^n u\,du = \frac{1}{n-1}\tan^{n-1} u - \int \tan^{n-2} u\,du$

31. $\int \cot^n u\,du = \frac{-1}{n-1}\cot^{n-1} u - \int \cot^{n-2} u\,du$

32. $\int \sec^n u\,du = \frac{1}{n-1}\tan u \sec^{n-2} u + \frac{n-2}{n-1}\int \sec^{n-2} u\,du$

33. $\int \csc^n u\,du = \frac{-1}{n-1}\cot u \csc^{n-2} u + \frac{n-2}{n-1}\int \csc^{n-2} u\,du$

34. $\int \sin au \sin bu\,du = \frac{\sin(a-b)u}{2(a-b)} - \frac{\sin(a+b)u}{2(a+b)} + C$

35. $\int \cos au \cos bu\,du = \frac{\sin(a-b)u}{2(a-b)} + \frac{\sin(a+b)u}{2(a+b)} + C$

36. $\int \sin au \cos bu\,du = -\frac{\cos(a-b)u}{2(a-b)} - \frac{\cos(a+b)u}{2(a+b)} + C$

37. $\int u \sin u\,du = \sin u - u\cos u + C$

38. $\int u \cos u\,du = \cos u + u\sin u + C$

39. $\int u^n \sin u\,du = -u^n \cos u + n\int u^{n-1}\cos u\,du$

40. $\int u^n \cos u\,du = u^n \sin u - n\int u^{n-1}\sin u\,du$

41. Erroneous nesting of equation structures

Exponential and Logarithmic Integrals

42. $\int ue^{au}\,du = \frac{1}{a^2}(au-1)e^{au} + C$

43. $\int u^n e^{au}\,du = \frac{1}{a}u^n e^{au} - \frac{n}{a}\int u^{n-1}e^{au}\,du$

44. $\int e^{au}\sin bu\,du = \frac{e^{au}}{a^2+b^2}(a\sin bu - b\cos bu) + C$

45. $\int e^{au}\cos bu\,du = \frac{e^{au}}{a^2+b^2}(a\cos bu + b\sin bu) + C$

46. $\int \ln u\,du = u\ln u - u + C$

47. $\int u^n \ln u\,du = \frac{u^{n+1}}{(n+1)^2}[(n+1)\ln u - 1] + C$

48. $\int \frac{1}{u\ln u}\,du = \ln|\ln u| + C$

Hyperbolic Integrals

49. $\int \sinh u\,du = \cosh u + C$

50. $\int \cosh u\,du = \sinh u + C$

51. $\int \tanh u\,du = \ln\cosh u + C$

52. $\int \coth u\,du = \ln|\sinh u| + C$

53. $\int \operatorname{sech} u\,du = \tan^{-1}|\sinh u| + C$

54. $\int \operatorname{csch} u\,du = \ln|\tanh \frac{1}{2}u| + C$

55. $\int \operatorname{sech}^2 u\,du = \tanh u + C$

56. $\int \operatorname{csch}^2 u\,du = -\coth u + C$

57. $\int \operatorname{sech} u \tanh u\,du = -\operatorname{sech} u + C$

58. $\int \operatorname{csch} u \coth u\,du = -\operatorname{csch} u + C$

Inverse Trigonometric Integrals

59. $\int \sin^{-1} u\,du = u\sin^{-1} u + \sqrt{1-u^2} + C$

60. $\int \cos^{-1} u\,du = u\cos^{-1} u - \sqrt{1-u^2} + C$

61. $\int \tan^{-1} u\,du = u\tan^{-1} u - \frac{1}{2}\ln(1+u^2) + C$

62. $\int u\sin^{-1} u\,du = \frac{2u^2-1}{4}\sin^{-1} u + \frac{u\sqrt{1-u^2}}{4} + C$

63. $\int u\cos^{-1} u\,du = \frac{2u^2-1}{4}\cos^{-1} u - \frac{u\sqrt{1-u^2}}{4} + C$

64. $\int u\tan^{-1} u\,du = \frac{u^2+1}{2}\tan^{-1} u - \frac{u}{2} + C$

65. $\int u^n \sin^{-1} u\,du = \frac{1}{n+1}\left[u^{n+1}\sin^{-1} u - \int \frac{u^{n+1}\,du}{\sqrt{1-u^2}}\right], \quad n \neq -1$

66. $\int u^n \cos^{-1} u\,du = \frac{1}{n+1}\left[u^{n+1}\cos^{-1} u + \int \frac{u^{n+1}\,du}{\sqrt{1-u^2}}\right], \quad n \neq -1$

67. $\int u^n \tan^{-1} u\,du = \frac{1}{n+1}\left[u^{n+1}\tan^{-1} u - \int \frac{u^{n+1}\,du}{1+u^2}\right], \quad n \neq -1$

Integrals Involving $a^2 + u^2$, $a > 0$

68. $\int \sqrt{a^2+u^2}\,du = \frac{u}{2}\sqrt{a^2+u^2} + \frac{a^2}{2}\ln\left(u + \sqrt{a^2+u^2}\right) + C$

69. $\int u^2\sqrt{a^2+u^2}\,du = \frac{u}{8}(a^2+2u^2)\sqrt{a^2+u^2} - \frac{a^4}{8}\ln\left(u + \sqrt{a^2+u^2}\right) + C$

70. $\int \frac{\sqrt{a^2+u^2}}{u}\,du = \sqrt{a^2+u^2} - a\ln\left|\frac{a+\sqrt{a^2+u^2}}{u}\right| + C$

71. $\int \frac{\sqrt{a^2+u^2}}{u^2}\,du = -\frac{\sqrt{a^2+u^2}}{u} + \ln\left(u + \sqrt{a^2+u^2}\right) + C$

72. $\int \frac{du}{\sqrt{a^2+u^2}} = \ln\left(u + \sqrt{a^2+u^2}\right) + C$

73. $\int \frac{u^2}{\sqrt{a^2+u^2}}\,du = \frac{u}{2}\left(\sqrt{a^2+u^2}\right) - \frac{a^2}{2}\ln\left(u + \sqrt{a^2+u^2}\right) + C$

74. $\int \frac{du}{u\sqrt{a^2+u^2}} = \frac{-1}{a}\ln\left|\frac{\sqrt{a^2+u^2}+a}{u}\right| + C$

75. $\int \frac{du}{u^2\sqrt{a^2+u^2}} = -\frac{\sqrt{a^2+u^2}}{a^2u} + C$

76. $\int \frac{du}{(a^2+u^2)^{3/2}} = \frac{u}{a^2\sqrt{a^2+u^2}} + C$

Integrals Involving $u^2 - a^2$, $a > 0$

77. $\displaystyle\int \sqrt{u^2-a^2}\,du = \frac{u}{2}\sqrt{u^2-a^2} - \frac{a^2}{2}\ln\left|u+\sqrt{u^2-a^2}\right| + C$

78. $\displaystyle\int u^2\sqrt{u^2-a^2}\,du = \frac{u}{8}(2u^2-a^2)\sqrt{u^2-a^2} - \frac{a^4}{8}\ln\left|u+\sqrt{u^2-a^2}\right| + C$

79. $\displaystyle\int \frac{\sqrt{u^2-a^2}}{u}\,du = \sqrt{u^2-a^2} - a\cos^{-1}\frac{a}{|u|} + C$

80. $\displaystyle\int \frac{\sqrt{u^2-a^2}}{u^2}\,du = -\frac{\sqrt{u^2-a^2}}{u} + \ln\left|u+\sqrt{u^2-a^2}\right| + C$

81. $\displaystyle\int \frac{du}{\sqrt{u^2-a^2}} = \ln\left|u+\sqrt{u^2-a^2}\right| + C$

82. $\displaystyle\int \frac{u^2}{\sqrt{u^2-a^2}}\,du = \frac{u}{2}\sqrt{u^2-a^2} + \frac{a^2}{2}\ln\left|u+\sqrt{u^2-a^2}\right| + C$

83. $\displaystyle\int \frac{du}{u^2\sqrt{u^2-a^2}} = \frac{\sqrt{u^2-a^2}}{a^2u} + C$

84. $\displaystyle\int \frac{du}{(u^2-a^2)^{3/2}} = -\frac{u}{a^2\sqrt{u^2-a^2}} + C$

Integrals Involving $a^2 - u^2$, $a > 0$

85. $\displaystyle\int \sqrt{a^2-u^2}\,du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\sin^{-1}\frac{u}{a} + C$

86. $\displaystyle\int u^2\sqrt{a^2-u^2}\,du = \frac{u}{8}(2u^2-a^2)\sqrt{a^2-u^2} + \frac{a^4}{8}\sin^{-1}\frac{u}{a} + C$

87. $\displaystyle\int \frac{\sqrt{a^2-u^2}}{u}\,du = \sqrt{a^2-u^2} - a\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right| + C$

88. $\displaystyle\int \frac{\sqrt{a^2-u^2}}{u^2}\,du = \frac{-1}{u}\sqrt{a^2-u^2} - \sin^{-1}\frac{u}{a} + C$

89. $\displaystyle\int \frac{u^2}{\sqrt{a^2-u^2}}\,du = \frac{1}{2}\left(-u\sqrt{a^2-u^2} + a^2\sin^{-1}\frac{u}{a}\right) + C$

90. $\displaystyle\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right| + C$

91. $\displaystyle\int \frac{du}{u^2\sqrt{a^2-u^2}} = -\frac{1}{a^2u}\sqrt{a^2-u^2} + C$

92. $\displaystyle\int \left(a^2-u^2\right)^{3/2}\,du = -\frac{u}{8}\left(2u^2-5a^2\right)\sqrt{a^2-u^2} + \frac{3a^4}{8}\sin^{-1}\frac{u}{a} + C$

93. $\displaystyle\int \frac{du}{(a^2-u^2)^{3/2}} = -\frac{u}{a^2\sqrt{a^2-u^2}} + C$

Integrals Involving $2au - u^2$, $a > 0$

94. $\displaystyle\int \sqrt{2au-u^2}\,du = \frac{u-a}{2}\sqrt{2au-u^2} + \frac{a^2}{2}\cos^{-1}\left(\frac{a-u}{a}\right) + C$

95. $\displaystyle\int \frac{du}{\sqrt{2au-u^2}} = \cos^{-1}\left(\frac{a-u}{a}\right) + C$

96. $\displaystyle\int u\sqrt{2au-u^2}\,du = \frac{2u^2-au-3a^2}{6}\sqrt{2au-u^2} + \frac{a^3}{2}\cos^{-1}\left(\frac{a-u}{a}\right) + C$

97. $\displaystyle\int \frac{du}{u\sqrt{2au-u^2}} = -\frac{\sqrt{2au-u^2}}{au} + C$

Integrals Involving $a + bu$, $a \neq 0$

98. $\displaystyle\int \frac{u}{a+bu}\,du = \frac{1}{b^2}(a+bu-a\ln|a+bu|) + C$

99. $\displaystyle\int \frac{u^2}{a+bu}\,du = \frac{1}{2b^3}\left[(a+bu)^2 - 4a(a+bu) + 2a^2\ln|a+bu|\right] + C$

100. $\displaystyle\int \frac{du}{u(a+bu)} = \frac{1}{a}\ln\left|\frac{u}{a+bu}\right| + C$

101. $\displaystyle\int \frac{du}{u^2(a+bu)} = -\frac{1}{au} + \frac{b}{a^2}\ln\left|\frac{a+bu}{u}\right| + C$

102. $\displaystyle\int \frac{u}{(a+bu)^2}\,du = \frac{a}{b^2(a+bu)} + \frac{1}{b^2}\ln|a+bu| + C$

103. $\displaystyle\int \frac{u}{u(a+bu)^2}\,du = \frac{1}{a(a+bu)} - \frac{1}{a^2}\ln\left|\frac{a+bu}{u}\right| + C$

104. $\displaystyle\int \frac{u^2}{(a+bu)^2}\,du = \frac{1}{b^3}\left(a+bu - \frac{a^2}{a+bu} - 2a\ln|a+bu|\right) + C$

105. $\displaystyle\int u\sqrt{a+bu}\,du = \frac{2}{15b^2}(3bu-2a)(a+bu)^{3/2} + C$

106. $\displaystyle\int \frac{u}{\sqrt{a+bu}}\,du = \frac{2}{3b^2}(bu-2a)\sqrt{a+bu} + C$

107. $\displaystyle\int \frac{u^2}{\sqrt{a+bu}}\,du = \frac{2}{15b^3}(8a^2+3b^2u^2-4abu)\sqrt{a+bu} + C$

108. $\displaystyle\int \frac{du}{u\sqrt{a+bu}} = \begin{cases} \frac{1}{\sqrt{a}}\ln\left|\frac{\sqrt{a+bu}-\sqrt{a}}{\sqrt{a+bu}+\sqrt{a}}\right| + C, & \text{if } a > 0 \\ \frac{\sqrt{2}}{\sqrt{-a}}\tan^{-1}\sqrt{\frac{a+bu}{-a}} + C, & \text{if } a < 0 \end{cases}$

109. $\displaystyle\int \frac{\sqrt{a+bu}}{u}\,du = 2\sqrt{a+bu} + a\int \frac{du}{u\sqrt{a+bu}}$

110. $\displaystyle\int \frac{\sqrt{a+bu}}{u^2}\,du = -\frac{\sqrt{a+bu}}{u} + \frac{b}{2}\int \frac{du}{u\sqrt{a+bu}}$

111. $\displaystyle\int u^n\sqrt{a+bu}\,du = \frac{2}{b(2n+3)}\left[u^n(a+bu)^{3/2} - na\int u^{n-1}\sqrt{a+bu}\,du\right]$

112. $\displaystyle\int \frac{u^n}{\sqrt{a+bu}}\,du = \frac{2u^n\sqrt{a+bu}}{b(2n+1)} - \frac{2na}{b(2n+1)}\int \frac{u^{n-1}}{\sqrt{a+bu}}\,du$

113. $\displaystyle\int \frac{du}{u^n\sqrt{a+bu}} = -\frac{\sqrt{a+bu}}{a(n-1)u^{n-1}} - \frac{b(2n-3)}{2a(n-1)}\int \frac{du}{u^{n-1}\sqrt{a+bu}}$

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INDEX

A

absolute convergence
9.5: Alternating Series

absolute error
7.6: Numerical Integration

absolute extremum
4.3: Maxima and Minima

absolute maximum
4.3: Maxima and Minima

absolute minimum
4.3: Maxima and Minima

Absolute Value Function
1.1: Review of Functions

acceleration
3.4: Derivatives as Rates of Change

acceleration vector
13.4: Motion in Space

air resistance
8.5: First-order Linear Equations

Airy's equation
10.4: Working with Taylor Series

algebraic function
1.2: Basic Classes of Functions

Alternating Harmonic Series
9.5: Alternating Series

alternating series
9.5: Alternating Series

Alternating Series Test
9.5: Alternating Series

amount of change
3.4: Derivatives as Rates of Change

angular coordinate
11.3: Polar Coordinates

angular frequency
17.3: Applications of Second-Order Differential Equations

antiderivative
4.10: Antiderivatives

arc length
6.4: Arc Length of a Curve and Surface Area

Arc length of a parametric curve
11.2: Calculus of Parametric Curves

Arc length of a polar curve
11.4: Area and Arc Length in Polar Coordinates

Archimedean spiral
11.3: Polar Coordinates

AREA BETWEEN TWO CURVES
6.1: Areas between Curves

AREA OF A PARALLELOGRAM
12.4: The Cross Product

Area of a region bounded by a polar curve
11.4: Area and Arc Length in Polar Coordinates

Area under a parametric curve
11.2: Calculus of Parametric Curves

Areas of Compound Regions
6.1: Areas between Curves

arithmetic sequence
9.1: Sequences

asymptotically stable solution
8.2: Direction Fields and Numerical Methods

asymptotically unstable solution
8.2: Direction Fields and Numerical Methods

autonomous differential equation
8.3: Separable Equations

average rate of change
3.4: Derivatives as Rates of Change

average value of a function
5.2: The Definite Integral

average velocity
2.1: A Preview of Calculus
3.1: Defining the Derivative

B

Bessel functions
17.4: Series Solutions of Differential Equations

binomial series
10.4: Working with Taylor Series

binormalbinormal vector
13.3: Arc Length and Curvature

boundary point
14.2: Limits and Continuity

C

Calculating flux
16.2: Line Integrals

cardioid
11.3: Polar Coordinates

carrying capacity
8.4: The Logistic Equation

catenary
6.9: Calculus of the Hyperbolic Functions

center of gravity
15.6: Calculating Centers of Mass and Moments of Inertia

center of mass
6.6: Moments and Centers of Mass
15.6: Calculating Centers of Mass and Moments of Inertia

chain rule
3.6: The Chain Rule

change of variables
5.5: Substitution

characteristic equation
17.1: Second-Order Linear Equations

circulation
16.4: Green's Theorem

Clairaut's theorem
14.3: Partial Derivatives
16.1: Vector Fields

closed curve
14.2: Limits and Continuity
16.3: Conservative Vector Fields

closed set
14.2: Limits and Continuity

Closeness
2.5: The Precise Definition of a Limit

Combining Vectors
12.1: Vectors in the Plane

comparison test
9.4: Comparison Tests

complementary equation
17.2: Nonhomogeneous Linear Equations

component
12.1: Vectors in the Plane

component functions
13.1: Vector-Valued Functions and Space Curves

composite function
1.1: Review of Functions

compound interest
6.8: Exponential Growth and Decay

computer algebra system
7.5: Other Strategies for Integration

concave down
4.5: Derivatives and the Shape of a Graph

concave up
4.5: Derivatives and the Shape of a Graph

concavity
4.5: Derivatives and the Shape of a Graph

concavity test
4.5: Derivatives and the Shape of a Graph

conditional convergence
9.5: Alternating Series

conic sections
11.5: Conic Sections
12.6: Quadric Surfaces

connected region
14.2: Limits and Continuity
16.3: Conservative Vector Fields

connected set
14.2: Limits and Continuity

conservative field
16.1: Vector Fields

Conservative Vector Fields
16.3: Conservative Vector Fields
constant multiple law for limits
2.3: The Limit Laws
constant multiple rule
3.3: Differentiation Rules
Constant Rule
3.3: Differentiation Rules
constraint
14.8: Lagrange Multipliers
continuity
2.4: Continuity
continuity at a point
2.4: Continuity
continuity from the left
2.4: Continuity
continuity from the right
2.4: Continuity
continuity over an interval
2.4: Continuity
contour map
14.1: Functions of Several Variables
14.3: Partial Derivatives
convergence of a series
9.2: Infinite Series
convergent sequence
9.1: Sequences
coordinate plane
12.2: Vectors in Three Dimensions
Cramer's Rule
17.2: Nonhomogeneous Linear Equations
critical point
4.3: Maxima and Minima
critical point of a function of two variables
14.7: Maxima/Minima Problems
critical points
14.7: Maxima/Minima Problems
cross product
12.4: The Cross Product
cubic function
1.2: Basic Classes of Functions
curl
16.5: Divergence and Curl
curl vector fields
16.7: Stokes' Theorem
curtate cycloid
11.1: Parametric Equations
curvature
13.3: Arc Length and Curvature
cusp
11.1: Parametric Equations
cusps
11.1: Parametric Equations
cycloid
11.1: Parametric Equations
Cycloids
11.1: Parametric Equations
cylinder
12.6: Quadric Surfaces
cylinders
12.6: Quadric Surfaces
cylindrical coordinate system
12.7: Cylindrical and Spherical Coordinates

D

decreasing on the interval I
1.1: Review of Functions
definite integral
5.2: The Definite Integral
degree
1.2: Basic Classes of Functions
delta ball
14.2: Limits and Continuity
delta disk
14.2: Limits and Continuity
DENSITY FUNCTION
6.5: Physical Applications of Integration
dependent variable
1.1: Review of Functions
derivative
3.1: Defining the Derivative
derivative function
3.2: The Derivative as a Function
Derivative of cosecant function
3.5: Derivatives of Trigonometric Functions
Derivative of cosine function
3.5: Derivatives of Trigonometric Functions
Derivative of cotangent function
3.5: Derivatives of Trigonometric Functions
Derivative of exponential functions
3.9: Derivatives of Exponential and Logarithmic Functions
Derivative of Hyperbolic Functions
6.9: Calculus of the Hyperbolic Functions
Derivative of inverse cosecant function
3.7: Derivatives of Inverse Functions
Derivative of inverse cosine function
3.7: Derivatives of Inverse Functions
Derivative of inverse cotangent function
3.7: Derivatives of Inverse Functions
Derivative of inverse secant function
3.7: Derivatives of Inverse Functions
Derivative of inverse tangent function
3.7: Derivatives of Inverse Functions
Derivative of logarithmic functions
3.9: Derivatives of Exponential and Logarithmic Functions
Derivative of parametric equations
11.2: Calculus of Parametric Curves
Derivative of secant function
3.5: Derivatives of Trigonometric Functions
Derivative of sine function
3.5: Derivatives of Trigonometric Functions
Derivative of tangent function
3.5: Derivatives of Trigonometric Functions
Derivative of the general exponential function
3.9: Derivatives of Exponential and Logarithmic Functions
Derivative of the general logarithmic function
3.9: Derivatives of Exponential and Logarithmic Functions
Derivative of the natural exponential function
3.9: Derivatives of Exponential and Logarithmic Functions
Derivative of the natural logarithmic function
3.9: Derivatives of Exponential and Logarithmic Functions
determinant
12.4: The Cross Product
difference law for limits
2.3: The Limit Laws
difference quotient
3.1: Defining the Derivative
Difference Rule
3.3: Differentiation Rules
Differentiability (three variables)
14.4: Tangent Planes and Linear Approximations
Differentiability (two variables)
14.4: Tangent Planes and Linear Approximations
differentiable
14.4: Tangent Planes and Linear Approximations
differentiable function
3.2: The Derivative as a Function
differentiable on S
3.2: The Derivative as a Function
differential
4.2: Linear Approximations and Differentials
differential calculus
2.1: A Preview of Calculus
differential equation
8.1: Basics of Differential Equations

differential form
4.2: Linear Approximations and Differentials
direction angles
12.3: The Dot Product
direction cosines
12.3: The Dot Product
direction vector
12.5: Equations of Lines and Planes in Space
DIRECTIONAL COSINES
14.6: Directional Derivatives and the Gradient
directional derivative
14.6: Directional Derivatives and the Gradient
directrix
11.5: Conic Sections
discontinuity at a point
2.4: Continuity
discriminant
11.5: Conic Sections
14.7: Maxima/Minima Problems
disk method
6.2: Determining Volumes by Slicing
Distance between a Plane and a Point
12.5: Equations of Lines and Planes in Space
divergence
16.5: Divergence and Curl
divergence of a series
9.2: Infinite Series
DIVERGENCE OF THE CURL
16.5: Divergence and Curl
Divergence Test
9.3: The Divergence and Integral Tests
divergence theorem
16.8: The Divergence Theorem
divergent sequence
9.1: Sequences
domain
1.1: Review of Functions
dot product
12.3: The Dot Product
double integral
15.1: Double Integrals over Rectangular Regions
double Riemann sum
15.1: Double Integrals over Rectangular Regions
doubling time
6.8: Exponential Growth and Decay

E

eccentricity
11.5: Conic Sections
ellipses
11.5: Conic Sections
ellipsoid
12.6: Quadric Surfaces
Elliptic cone
12.6: Quadric Surfaces
Elliptic paraboloid
12.6: Quadric Surfaces
end behavior
4.6: Limits at Infinity and Asymptotes
equilibrium solution
8.2: Direction Fields and Numerical Methods
equivalent vectors
12.1: Vectors in the Plane
Euler’s Method
8.2: Direction Fields and Numerical Methods
even function
1.1: Review of Functions
exponent
1.5: Exponential and Logarithmic Functions
exponential function
6.7: Integrals, Exponential Functions, and Logarithms
extreme value theorem
4.3: Maxima and Minima

F

family of solutions
8.1: Basics of Differential Equations
Faraday's Law
16.7: Stokes’ Theorem
Fermat’s theorem
4.3: Maxima and Minima
Fermat’s Theorem for Functions of Two Variables
14.7: Maxima/Minima Problems
Fibonacci Numbers
9.1: Sequences
First Derivative Test
4.5: Derivatives and the Shape of a Graph
first moment
6.6: Moments and Centers of Mass
flux
16.2: Line Integrals
16.4: Green’s Theorem
16.8: The Divergence Theorem
flux integral
16.6: Surface Integrals
focal parameter
11.5: Conic Sections
focus
11.5: Conic Sections
Forced harmonic motion
17.3: Applications of Second-Order Differential Equations
Frenet frame of reference
13.3: Arc Length and Curvature
frustum
6.4: Arc Length of a Curve and Surface Area
FUBINI’S THEOREM
15.5: Triple Integrals in Cylindrical and Spherical Coordinates
Fubini’s thereom
15.4: Triple Integrals
Fubini's theorem
15.1: Double Integrals over Rectangular Regions
function
1.1: Review of Functions
Function of Two Variables
14.1: Functions of Several Variables
FUNDAMENTAL THEOREM FOR LINE INTEGRALS
16.3: Conservative Vector Fields
16.8: The Divergence Theorem
fundamental theorem of calculus
5.3: The Fundamental Theorem of Calculus
16.8: The Divergence Theorem
fundamental theorem of calculus, part 1
5.3: The Fundamental Theorem of Calculus
fundamental theorem of calculus, part 2
5.3: The Fundamental Theorem of Calculus

G

Gauss' Law
16.8: The Divergence Theorem
general form
11.5: Conic Sections
general form of the equation of a plane
12.5: Equations of Lines and Planes in Space
general solution
8.1: Basics of Differential Equations
generalized chain rule
14.5: The Chain Rule for Multivariable Functions
Geometric Sequences
9.1: Sequences
geometric series
9.2: Infinite Series
9.5: Alternating Series
gradient
14.6: Directional Derivatives and the Gradient
gradient field
16.1: Vector Fields
graph of a function
1.1: Review of Functions
graph of a function of two variables
14.1: Functions of Several Variables
Green's Theorem
16.4: Green’s Theorem
16.8: The Divergence Theorem

grid curves
16.6: Surface Integrals
growth rate
8.4: The Logistic Equation

H

harmonic series
9.2: Infinite Series
heat flow
16.6: Surface Integrals
helix
13.1: Vector-Valued Functions and Space Curves
Hooke's law
6.5: Physical Applications of Integration
17.3: Applications of Second-Order Differential Equations
horizontal asymptote
4.6: Limits at Infinity and Asymptotes
horizontal line test
1.4: Inverse Functions
hydrostatic pressure
6.5: Physical Applications of Integration
hyperbolas
11.5: Conic Sections
hyperbolic functions
1.5: Exponential and Logarithmic Functions
hyperboloid
12.6: Quadric Surfaces
Hyperboloid of one sheet
12.6: Quadric Surfaces
Hyperboloid of two sheets
12.6: Quadric Surfaces
hypocycloid
11.1: Parametric Equations

I

implicit differentiation
3.8: Implicit Differentiation
improper double integral
15.2: Double Integrals over General Regions
improper integral
7.7: Improper Integrals
increasing on the interval I
1.1: Review of Functions
indefinite integral
4.10: Antiderivatives
independent variable
1.1: Review of Functions
indeterminate forms
4.8: L'Hôpital's Rule
index variable
9.1: Sequences
infinite discontinuity
2.4: Continuity
infinite limit
2.2: The Limit of a Function
infinite limit at infinity
4.6: Limits at Infinity and Asymptotes
infinite series
9.2: Infinite Series
inflection point
4.5: Derivatives and the Shape of a Graph
initial point
12.1: Vectors in the Plane
initial population
8.4: The Logistic Equation
initial value
8.1: Basics of Differential Equations
initial value problem
4.10: Antiderivatives
initial velocity
8.1: Basics of Differential Equations
inner product
12.3: The Dot Product
instantaneous rate of change
3.1: Defining the Derivative
instantaneous velocity
2.1: A Preview of Calculus
3.1: Defining the Derivative
integrable function
5.2: The Definite Integral
integral calculus
2.1: A Preview of Calculus
Integral Test
9.3: The Divergence and Integral Tests
Integrals of Exponential Functions
5.6: Integrals Involving Exponential and Logarithmic Functions
Integrals That Produce Inverse Trigonometric Functions
5.7: Integrals Resulting in Inverse Trigonometric Functions
5.7: Integrals Resulting in Inverse Trigonometric Functions
integrand
5.2: The Definite Integral
integrating factor
8.5: First-order Linear Equations
integration by parts
7.1: Integration by Parts
integration by substitution
5.5: Substitution
Integration Formulas Involving Logarithmic Functions
5.6: Integrals Involving Exponential and Logarithmic Functions
integration table
7.5: Other Strategies for Integration
interior point
14.2: Limits and Continuity
Intermediate Value Theorem
2.4: Continuity
intermediate variable
14.5: The Chain Rule for Multivariable Functions
Interpretation of Curl
16.7: Stokes' Theorem
interval of convergence
10.1: Power Series and Functions
intuitive definition of the limit
2.2: The Limit of a Function
inverse function
1.4: Inverse Functions
inverse function theorem
3.7: Derivatives of Inverse Functions
inverse hyperbolic functions
1.5: Exponential and Logarithmic Functions
inverse trigonometric functions
1.4: Inverse Functions
iterated integral
15.1: Double Integrals over Rectangular Regions
iterative process
4.9: Newton's Method

J

Jacobian
15.7: Change of Variables in Multiple Integrals
jump discontinuity
2.4: Continuity

K

Kepler's laws of planetary motion
13.4: Motion in Space

L

L'Hôpital's rule
4.8: L'Hôpital's Rule
Lagrange multiplier
14.8: Lagrange Multipliers
level curve of a function of two variables
14.1: Functions of Several Variables
level surface of a function of three variables
14.1: Functions of Several Variables
limacon
11.3: Polar Coordinates
limit
2.1: A Preview of Calculus
limit at infinity
4.6: Limits at Infinity and Asymptotes

Limit Comparison Test
9.4: Comparison Tests
limit laws
2.3: The Limit Laws
LIMIT LAWS FOR FUNCTIONS OF TWO VARIABLES
14.2: Limits and Continuity
limit of a sequence
9.1: Sequences
line integrals
16.2: Line Integrals
linear
8.5: First-order Linear Equations
linear approximation
4.2: Linear Approximations and Differentials
14.4: Tangent Planes and Linear Approximations
linear function
1.2: Basic Classes of Functions
linearization
4.2: Linear Approximations and Differentials
linearly dependent
17.1: Second-Order Linear Equations
linearly independent
17.1: Second-Order Linear Equations
local extremum
4.3: Maxima and Minima
local maximum
4.3: Maxima and Minima
local minimum
4.3: Maxima and Minima
Logarithmic Function
1.2: Basic Classes of Functions
logistic differential equation
8.4: The Logistic Equation
lower sum
5.1: Approximating Areas

M

Möbius strip
16.6: Surface Integrals
Maclaurin polynomial
10.3: Taylor and Maclaurin Series
Maclaurin series
10.3: Taylor and Maclaurin Series
magnitude
12.1: Vectors in the Plane
major axis
11.5: Conic Sections
marginal cost
3.4: Derivatives as Rates of Change
marginal profit
3.4: Derivatives as Rates of Change
marginal revenue
3.4: Derivatives as Rates of Change
mass flux
16.6: Surface Integrals
mathematical model
1.2: Basic Classes of Functions
Mean Value Theorem
4.4: The Mean Value Theorem
Mean Value Theorem for Integrals
5.3: The Fundamental Theorem of Calculus
method of cylindrical shells
6.3: Volumes of Revolution - Cylindrical Shells
method of equating coefficients
7.4: Partial Fractions
method of exhaustion
5.1: Approximating Areas
method of Lagrange multipliers
14.8: Lagrange Multipliers
method of strategic substitution
7.4: Partial Fractions
method of undetermined coefficients
17.2: Nonhomogeneous Linear Equations
method of variation of parameters
17.2: Nonhomogeneous Linear Equations
midpoint rule
7.6: Numerical Integration
minor axis
11.5: Conic Sections
mixed partial derivatives
14.3: Partial Derivatives
Moments of Inertia
6.6: Moments and Centers of Mass
15.6: Calculating Centers of Mass and Moments of Inertia
Monotone Convergence Theorem
9.1: Sequences
monotone sequence
9.1: Sequences
Multiple Integration
15: Multiple Integration
multivariable calculus
2.1: A Preview of Calculus

N

nappe
11.5: Conic Sections
nappes
11.5: Conic Sections
natural exponential function
1.5: Exponential and Logarithmic Functions
natural logarithm
1.5: Exponential and Logarithmic Functions
Natural logarithm function
6.7: Integrals, Exponential Functions, and Logarithms
net change theorem
5.4: Integration Formulas and the Net Change Theorem
net signed area
5.2: The Definite Integral
Newton's Law of Cooling
6.8: Exponential Growth and Decay
8.3: Separable Equations
Newton's method
4.9: Newton's Method
nonelementary integral
10.4: Working with Taylor Series
nonhomogeneous linear equation
17.1: Second-Order Linear Equations
normal component of acceleration
13.4: Motion in Space
normal plane
13.3: Arc Length and Curvature
normal vector
12.5: Equations of Lines and Planes in Space
Normalization
12.1: Vectors in the Plane
number e
1.5: Exponential and Logarithmic Functions
numerical integration
7.6: Numerical Integration

O

objective function
14.8: Lagrange Multipliers
oblique asymptote
4.6: Limits at Infinity and Asymptotes
octants
12.2: Vectors in Three Dimensions
odd function
1.1: Review of Functions
open set
14.2: Limits and Continuity
optimization problem
14.8: Lagrange Multipliers
optimization problems
4.7: Applied Optimization Problems
order of a differential equation
8.1: Basics of Differential Equations
ordinary point
17.4: Series Solutions of Differential Equations
orientation
11.1: Parametric Equations
orientation of a surface
16.6: Surface Integrals

orthogonal vectors
12.3: The Dot Product
osculating circle
13.3: Arc Length and Curvature
osculating plane
13.3: Arc Length and Curvature

P

parabola
11.5: Conic Sections
paraboloid
12.6: Quadric Surfaces
parallelepiped
12.4: The Cross Product
parallelogram method
12.1: Vectors in the Plane
parameter
11.1: Parametric Equations
parameter domain
16.6: Surface Integrals
parameter space
16.6: Surface Integrals
parameterization of a curve
11.1: Parametric Equations
parameterized surface
16.6: Surface Integrals
parametric curve
11.1: Parametric Equations
parametric equations
11.1: Parametric Equations
parametric equations of a line
12.5: Equations of Lines and Planes in Space
parametric surface
16.6: Surface Integrals
Parametric Surfaces
16.6: Surface Integrals
partial derivative
14.3: Partial Derivatives
partial differential equation
14.3: Partial Derivatives
Partial Differentiation
14.3: Partial Derivatives
partial fraction decomposition
7.4: Partial Fractions
partial sum
9.2: Infinite Series
particular solution
8.1: Basics of Differential Equations
17.2: Nonhomogeneous Linear Equations
partition
5.1: Approximating Areas
percentage error
4.2: Linear Approximations and Differentials
periodic function
1.3: Trigonometric Functions
phase line
8.4: The Logistic Equation
planar transformation
15.7: Change of Variables in Multiple Integrals
plane curve
13.1: Vector-Valued Functions and Space Curves
points of inflection
4.5: Derivatives and the Shape of a Graph
Polar Areas
15.3: Double Integrals in Polar Coordinates
polar axis
11.3: Polar Coordinates
polar coordinate system
11.3: Polar Coordinates
polar equation
11.3: Polar Coordinates
polar rectangle
15.3: Double Integrals in Polar Coordinates
Polar Volumes
15.3: Double Integrals in Polar Coordinates
pole
11.3: Polar Coordinates
polynomial function
1.2: Basic Classes of Functions
population growth
6.8: Exponential Growth and Decay
population growth rate
3.4: Derivatives as Rates of Change
power function
1.2: Basic Classes of Functions
Power Reduction Formula
7.2: Trigonometric Integrals
power rule
3.3: Differentiation Rules
Power rule with rational exponents
3.7: Derivatives of Inverse Functions
power series
10: Power Series
10.1: Power Series and Functions
principal unit normal vector
13.3: Arc Length and Curvature
principal unit tangent vector
13.2: Calculus of Vector-Valued Functions
product law for limits
2.3: The Limit Laws
product rule
3.3: Differentiation Rules
projectile motion
13.4: Motion in Space
propagated error
4.2: Linear Approximations and Differentials

Q

quadratic function
1.2: Basic Classes of Functions
Quadric Surfaces
12.6: Quadric Surfaces
quotient law for limits
2.3: The Limit Laws
quotient rule
3.3: Differentiation Rules

R

radial coordinate
11.3: Polar Coordinates
radial density
6.5: Physical Applications of Integration
radians
1.3: Trigonometric Functions
radius of convergence
10.1: Power Series and Functions
radius of curvature
13.3: Arc Length and Curvature
radius of gyration
15.6: Calculating Centers of Mass and Moments of Inertia
range
1.1: Review of Functions
ratio test
9.6: Ratio and Root Tests
rational function
1.2: Basic Classes of Functions
recurrence relation
9.1: Sequences
region
14.2: Limits and Continuity
regular parameterization
16.6: Surface Integrals
regular partition
5.1: Approximating Areas
related rates
4.1: Related Rates
relative error
4.2: Linear Approximations and Differentials
7.6: Numerical Integration
Remainder Estimate
9.3: The Divergence and Integral Tests
removable discontinuity
2.4: Continuity

reparameterization
13.1: Vector-Valued Functions and Space Curves
repeated linear factors
7.4: Partial Fractions
restricted domain
1.4: Inverse Functions
Riemann Rearrangement Theorem
9.5: Alternating Series
Riemann sum
5.1: Approximating Areas
RLC series circuit
17.3: Applications of Second-Order Differential Equations
Rolle's Theorem
4.4: The Mean Value Theorem
root function
1.2: Basic Classes of Functions
root law for limits
2.3: The Limit Laws
root test
9.6: Ratio and Root Tests
rose
11.3: Polar Coordinates
rotational field
16.1: Vector Fields
rulings
12.6: Quadric Surfaces

S

saddle point
14.7: Maxima/Minima Problems
scalar
12.1: Vectors in the Plane
Scalar Equation of a Plane
12.5: Equations of Lines and Planes in Space
scalar line integral
16.2: Line Integrals
scalar multiplication
12.1: Vectors in the Plane
scalar product
12.3: The Dot Product
scalar projection
12.3: The Dot Product
secant line
2.1: A Preview of Calculus
second derivative test
4.5: Derivatives and the Shape of a Graph
separation of variables
8.3: Separable Equations
sequence
9.1: Sequences
sequences
9.1: Sequences
sigma notation
5.1: Approximating Areas
simple harmonic motion
17.3: Applications of Second-Order Differential Equations
singular point
17.4: Series Solutions of Differential Equations
skew lines
12.5: Equations of Lines and Planes in Space
slicing method
6.2: Determining Volumes by Slicing
slope
1.2: Basic Classes of Functions
slope field
8.2: Direction Fields and Numerical Methods
smooth
13.3: Arc Length and Curvature
solid of revolution
6.2: Determining Volumes by Slicing
6.3: Volumes of Revolution - Cylindrical Shells
solution curve
8.2: Direction Fields and Numerical Methods
solution to a differential equation
8.1: Basics of Differential Equations
space curve
13.1: Vector-Valued Functions and Space Curves
speed
3.4: Derivatives as Rates of Change
sphere
12.2: Vectors in Three Dimensions
spherical coordinate system
12.7: Cylindrical and Spherical Coordinates
spherical coordinates
12.7: Cylindrical and Spherical Coordinates
squeeze theorem
2.3: The Limit Laws
Squeeze Theorem for Sequences
9.1: Sequences
standard equation of a sphere
12.2: Vectors in Three Dimensions
standard form
11.5: Conic Sections
standard unit vectors
12.1: Vectors in the Plane
step size
8.2: Direction Fields and Numerical Methods
Stokes' Theorem
16.7: Stokes' Theorem
16.8: The Divergence Theorem
sum law for limits
2.3: The Limit Laws
Sum Rule
3.3: Differentiation Rules
summation notation
5.1: Approximating Areas
superposition principle
17.1: Second-Order Linear Equations
surface
14.1: Functions of Several Variables
surface area
6.4: Arc Length of a Curve and Surface Area
16.6: Surface Integrals
Surface area generated by a parametric curve
11.2: Calculus of Parametric Curves
surface independent
16.7: Stokes' Theorem
surface integral
16.6: Surface Integrals
surface integral of a vector field
16.6: Surface Integrals
surface integrals
16.6: Surface Integrals
surface of revolution
6.4: Arc Length of a Curve and Surface Area
Symmetric Equations of a Line
12.5: Equations of Lines and Planes in Space
symmetry about the origin
1.1: Review of Functions

T

table of values
1.1: Review of Functions
tangent
2.1: A Preview of Calculus
tangent line
3.1: Defining the Derivative
tangent line approximation
4.2: Linear Approximations and Differentials
tangent plane
14.4: Tangent Planes and Linear Approximations
tangent vector
13.2: Calculus of Vector-Valued Functions
tangential component of acceleration
13.4: Motion in Space
Taylor polynomials
10.3: Taylor and Maclaurin Series
Taylor series
10.3: Taylor and Maclaurin Series
Taylor's theorem with remainder
10.3: Taylor and Maclaurin Series

telescoping series
9.2: Infinite Series
term
9.1: Sequences
terminal point
12.1: Vectors in the Plane
The Dot Product
12.3: The Dot Product
The Logistic Equation
8.4: The Logistic Equation
The power rule for functions
3.6: The Chain Rule
Theorem of Pappus
6.6: Moments and Centers of Mass
threshold population
8.4: The Logistic Equation
torque
12.4: The Cross Product
total area
5.2: The Definite Integral
total differential
14.4: Tangent Planes and Linear Approximations
trace
12.6: Quadric Surfaces
transcendental function
1.2: Basic Classes of Functions
transformation
15.7: Change of Variables in Multiple Integrals
transformation of a function
1.2: Basic Classes of Functions
Triangle Inequality
12.1: Vectors in the Plane
triangle method
12.1: Vectors in the Plane
trigonometric functions
1.3: Trigonometric Functions
trigonometric identity
1.3: Trigonometric Functions
trigonometric integral
7.2: Trigonometric Integrals
trigonometric substitution
7.3: Trigonometric Substitution
triple integral
15.4: Triple Integrals
triple integral in cylindrical coordinates
15.5: Triple Integrals in Cylindrical and Spherical Coordinates
triple integral in spherical coordinates
15.5: Triple Integrals in Cylindrical and Spherical Coordinates
Triple Scalar Product
12.4: The Cross Product
Type I
15.2: Double Integrals over General Regions
Type II
15.2: Double Integrals over General Regions

U

unit circle
1.3: Trigonometric Functions
unit vector
12.1: Vectors in the Plane
upper sum
5.1: Approximating Areas

V

variable of integration
5.2: The Definite Integral
vector
12.1: Vectors in the Plane
vector addition
12.1: Vectors in the Plane
vector difference
12.1: Vectors in the Plane
vector equation of a line
12.5: Equations of Lines and Planes in Space
Vector Equation of a Plane
12.5: Equations of Lines and Planes in Space
vector fields
16.1: Vector Fields
VECTOR LINE INTEGRALS
16.2: Line Integrals
vector parameterization
13.1: Vector-Valued Functions and Space Curves
vector product
12.4: The Cross Product
vector projection
12.3: The Dot Product
vector sum
12.1: Vectors in the Plane
Vectors in R3
12.2: Vectors in Three Dimensions
velocity vector
13.4: Motion in Space
vertex
11.5: Conic Sections
vertical asymptote
2.2: The Limit of a Function
vertical line test
1.1: Review of Functions
vertical trace
14.1: Functions of Several Variables
Volume by Shells
6.3: Volumes of Revolution - Cylindrical Shells
12.7: Cylindrical and Spherical Coordinates

W

washer method
6.2: Determining Volumes by Slicing
Witch of Agnesi
11.1: Parametric Equations
work
6.5: Physical Applications of Integration
work done by a force
12.3: The Dot Product

Z

zero vector
12.1: Vectors in the Plane
zeros of a function
1.1: Review of Functions

Glossary

absolute convergence | if the series $\sum_{n=1}^{\infty}|a_n|$ converges, the series $\sum_{n=1}^{\infty}a_n$ is said to converge absolutely

absolute error | if B is an estimate of some quantity having an actual value of A, then the absolute error is given by $|A-B|$

absolute extremum | if f has an absolute maximum or absolute minimum at c, we say f has an absolute extremum at c

absolute maximum | if $f(c)\geq f(x)$ for all x in the domain of f, we say f has an absolute maximum at c

absolute minimum | if $f(c)\leq f(x)$ for all x in the domain of f, we say f has an absolute minimum at c

absolute value function | $f(x)=\{-x$, if $x<0$ \if $x\geq 0$

acceleration | is the rate of change of the velocity, that is, the derivative of velocity

acceleration vector | the second derivative of the position vector

algebraic function | a function involving any combination of only the basic operations of addition, subtraction, multiplication, division, powers, and roots applied to an input variable x

alternating series | a series of the form $\sum_{n=1}^{\infty}(-1)^{n+1}b_n$ or $\sum_{n=1}^{\infty}(-1)^{n}b_n$, where $b_n\geq 0$, is called an alternating series

alternating series test | for an alternating series of either form, if $b_{n+1}\leq b_n$ for all integers $n\geq 1$ and $b_n\to 0$, then an alternating series converges

amount of change | the amount of a function $f(x)$ over an interval $[x,x+h]isf(x+h)-f(x)$

angular coordinate | θ the angle formed by a line segment connecting the origin to a point in the polar coordinate system with the positive radial (x) axis, measured counterclockwise

antiderivative | a function F such that $F'(x)=f(x)$ for all x in the domain of f is an antiderivative of f

arc length | the arc length of a curve can be thought of as the distance a person would travel along the path of the curve

arc-length function | a function $s(t)$ that describes the arc length of curve C as a function of t

arc-length parameterization | a reparameterization of a vector-valued function in which the parameter is equal to the arc length

arithmetic sequence | a sequence in which the difference between every pair of consecutive terms is the same is called an arithmetic sequence

asymptotically semi-stable solution | $y=k$ if it is neither asymptotically stable nor asymptotically unstable

asymptotically stable solution | $y=k$ if there exists $\varepsilon>0$ such that for any value $c\in(k-\varepsilon,k+\varepsilon)$ the solution to the initial-value problem $y'=f(x,y),y(x_0)=c$ approaches k as x approaches infinity

asymptotically unstable solution | $y=k$ if there exists $\varepsilon>0$ such that for any value $c\in(k-\varepsilon,k+\varepsilon)$ the solution to the initial-value problem $y'=f(x,y),y(x_0)=c$ never approaches k as $x$approaches infinity

autonomous differential equation | an equation in which the right-hand side is a function of y alone

average rate of change | is a function $f(x)$ over an interval $[x,x+h]$ is $\frac{f(x+h)-f(a)}{b-a}$

average value of a function | (or f_{ave}) the average value of a function on an interval can be found by calculating the definite integral of the function and dividing that value by the length of the interval

average velocity | the change in an object's position divided by the length of a time period; the average velocity of an object over a time interval $[t,a]$ (if $t<a$ or $[a,t]$ if $t>a$), with a position given by $s(t)$, that is $v_{ave}=\dfrac{s(t)-s(a)}{t-a}$

base | the number b in the exponential function $f(x)=b^x$ and the logarithmic function $f(x)=\log_b x$

binomial series | the Maclaurin series for $f(x)=(1+x)^r$; it is given by $(1+x)^r=\sum_{n=0}^{\infty}\binom{r}{n}x^n=1+rx+\dfrac{r(r-1)}{2!}x^2+\cdots+\dfrac{r(r-1)\cdots(r-n+1)}{n!}x^n+\cdots$ for $|x|<1$

binormal vector | a unit vector orthogonal to the unit tangent vector and the unit normal vector

boundary conditions | the conditions that give the state of a system at different times, such as the position of a spring-mass system at two different times

boundary point | a point P_0 of R is a boundary point if every δ disk centered around P_0 contains points both inside and outside R

boundary-value problem | a differential equation with associated boundary conditions

bounded above | a sequence a_n is bounded above if there exists a constant M such that $a_n\leq M$ for all positive integers n

bounded below | a sequence a_n is bounded below if there exists a constant M such that $M\leq a_n$ for all positive integers n

bounded sequence | a sequence a_n is bounded if there exists a constant M such that $|a_n|\leq M$ for all positive integers n

cardioid | a plane curve traced by a point on the perimeter of a circle that is rolling around a fixed circle of the same radius; the equation of a cardioid is $r=a(1+\sin\theta)$ or $r=a(1+\cos\theta)$

carrying capacity | the maximum population of an organism that the environment can sustain indefinitely

catenary | a curve in the shape of the function $y=a\cdot\cosh(x/a)$ is a catenary; a cable of uniform density suspended between two supports assumes the shape of a catenary

center of mass | the point at which the total mass of the system could be concentrated without changing the moment

centroid | the centroid of a region is the geometric center of the region; laminas are often represented by regions in the plane; if the lamina has a constant density, the center of mass of the lamina depends only on the shape of the corresponding planar region; in this case, the center of mass of the lamina corresponds to the centroid of the representative region

chain rule | the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function

change of variables | the substitution of a variable, such as u, for an expression in the integrand

characteristic equation | the equation $a\lambda^2 + b\lambda + c = 0$ for the differential equation $ay'' + by' + cy = 0$

circulation | the tendency of a fluid to move in the direction of curve C. If C is a closed curve, then the circulation of $\vec{\mathbf{F}}$ along C is line integral $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$, which we also denote $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$.

closed curve | a curve for which there exists a parameterization $\vec{\mathbf{r}}(t), a \le t \le b$, such that $\vec{\mathbf{r}}(a) = \vec{\mathbf{r}}(b)$, and the curve is traversed exactly once

closed curve | a curve that begins and ends at the same point

closed set | a set S that contains all its boundary points

comparison test | If $0 \le a_n \le b_n$ for all $n \ge N$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges; if $a_n \ge b_n \ge 0$ for all $n \ge N$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.

complementary equation | for the nonhomogeneous linear differential equation

$$a + 2(x)y'' + a_1(x)y' + a_0(x)y = r(x),$$

the associated homogeneous equation, called the *complementary equation*, is

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

component | a scalar that describes either the vertical or horizontal direction of a vector

component functions | the component functions of the vector-valued function $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ are $f(t)$ and $g(t)$, and the component functions of the vector-valued function $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$ are $f(t)$, $g(t)$ and $h(t)$

composite function | given two functions f and g, a new function, denoted $g \circ f$, such that $(g \circ f)(x) = g(f(x))$

computer algebra system (CAS) | technology used to perform many mathematical tasks, including integration

concave down | if f is differentiable over an interval I and f' is decreasing over I, then f is concave down over I

concave up | if f is differentiable over an interval I and f' is increasing over I, then f is concave up over I

concavity | the upward or downward curve of the graph of a function

concavity test | suppose f is twice differentiable over an interval I; if $f'' > 0$ over I, then f is concave up over I; if $f'' <$ over I, then f is concave down over I

conditional convergence | if the series $\sum_{n=1}^{\infty} a_n$ converges, but the series $\sum_{n=1}^{\infty} |a_n|$ diverges, the series $\sum_{n=1}^{\infty} a_n$ is said to converge conditionally

conic section | a conic section is any curve formed by the intersection of a plane with a cone of two nappes

connected region | a region in which any two points can be connected by a path with a trace contained entirely inside the region

connected set | an open set S that cannot be represented as the union of two or more disjoint, nonempty open subsets

conservative field | a vector field for which there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$

constant multiple law for limits | the limit law

$$\lim_{x \to a} cf(x) = c \cdot \lim_{x \to a} f(x) = cL$$

constant multiple rule | the derivative of a constant c multiplied by a function f is the same as the constant multiplied by the derivative: $\frac{d}{dx}\big(cf(x)\big) = cf'(x)$

constant rule | the derivative of a constant function is zero: $\frac{d}{dx}(c) = 0$, where c is a constant

constraint | an inequality or equation involving one or more variables that is used in an optimization problem; the constraint enforces a limit on the possible solutions for the problem

continuity at a point | A function $f(x)$ is continuous at a point a if and only if the following three conditions are satisfied: (1) $f(a)$ is defined, (2) $\lim_{x \to a} f(x)$ exists, and (3) $\lim x \to af(x) = f(a)$

continuity from the left | A function is continuous from the left at b if $\lim_{x \to b^-} f(x) = f(b)$

continuity from the right | A function is continuous from the right at a if $\lim_{x \to a^+} f(x) = f(a)$

continuity over an interval | a function that can be traced with a pencil without lifting the pencil; a function is continuous over an open interval if it is continuous at every point in the interval; a function $f(x)$ is continuous over a closed interval of the form $[a, b]$ if it is continuous at every point in (a, b), and it is continuous from the right at a and from the left at b

contour map | a plot of the various level curves of a given function $f(x, y)$

convergence of a series | a series converges if the sequence of partial sums for that series converges

convergent sequence | a convergent sequence is a sequence a_n for which there exists a real number L such that a_n is arbitrarily close to L as long as n is sufficiently large

coordinate plane | a plane containing two of the three coordinate axes in the three-dimensional coordinate system, named by the axes it contains: the xy-plane, xz-plane, or the yz-plane

critical point | if $f'(c) = 0$ or $f'(c)$ is undefined, we say that c is a critical point of f

critical point of a function of two variables | the point (x_0, y_0) is called a critical point of $f(x, y)$ if one of the two following conditions holds: 1. $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$ 2. At least one of $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ do not exist

cross product | $\vec{\mathbf{u}} \times \vec{\mathbf{v}} = (u_2v_3 - u_3v_2)\hat{\mathbf{i}} - (u_1v_3 - u_3v_1)\hat{\mathbf{j}} + (u_1v_2 - u_2v_1)\hat{\mathbf{k}}$, where $\vec{\mathbf{u}} = \langle u_1, u_2, u_3\rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3\rangle$ determinant a real number associated with a square matrix parallelepiped a three-dimensional prism with six faces that are parallelograms torque the effect of a force that causes an object to rotate triple scalar product the dot product of a vector with the cross product of two other vectors: $\vec{\mathbf{u}} \cdot (\vec{\mathbf{v}} \times \vec{\mathbf{w}})$ vector product the cross product of two vectors.

cross-section | the intersection of a plane and a solid object

cubic function | a polynomial of degree 3; that is, a function of the form $f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$

curl | the curl of vector field $\vec{\mathbf{F}} = \langle P, Q, R\rangle$, denoted $\vec{\nabla} \times \vec{\mathbf{F}}$ is the "determinant" of the matrix

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

and is given by the expression $(R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}}$; it measures the tendency of particles at a point to rotate about the axis that points in the direction of the curl at the point

curvature | the derivative of the unit tangent vector with respect to the arc-length parameter

cusp | a pointed end or part where two curves meet

cycloid | the curve traced by a point on the rim of a circular wheel as the wheel rolls along a straight line without slippage

cylinder | a set of lines parallel to a given line passing through a given curve

cylindrical coordinate system | a way to describe a location in space with an ordered triple (r, θ, z), where (r, θ) represents the polar coordinates of the point's projection in the xy-plane, and z represents the point's projection onto the z-axis

decreasing on the interval

definite integral | a primary operation of calculus; the area between the curve and the x-axis over a given interval is a definite integral

definite integral of a vector-valued function | the vector obtained by calculating the definite integral of each of the component functions of a given vector-valued function, then using the results as the components of the resulting function

degree | for a polynomial function, the value of the largest exponent of any term

density function | a density function describes how mass is distributed throughout an object; it can be a linear density, expressed in terms of mass per unit length; an area density, expressed in terms of mass per unit area; or a volume density, expressed in terms of mass per unit volume; weight-density is also used to describe weight (rather than mass) per unit volume

dependent variable | the output variable for a function

derivative | the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative

derivative function | gives the derivative of a function at each point in the domain of the original function for which the derivative is defined

derivative of a vector-valued function | the derivative of a vector-valued function $\vec{\mathbf{r}}(t)$ is $\vec{\mathbf{r}}'(t) = \lim_{\Delta t \to 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t}$, provided the limit exists

difference law for limits | the limit law

$$\begin{aligned} &\lim_{x \to a}(f(x) \\ &- g(x)) \\ &= \lim_{x \to a} f(x) \\ &- \lim_{x \to a} g(x) \\ &= L - M \end{aligned}$$

difference quotient | of a function $f(x)$ at a is given by $\frac{f(a+h) - f(a)}{h}$ or $\frac{f(x) - f(a)}{x - a}$

difference rule | the derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g: $\frac{d}{dx}\big(f(x) - g(x)\big) = f'(x) - g'(x)$

differentiable | a function $f(x, y)$ is differentiable at (x_0, y_0) if $f(x, y)$ can be expressed in the form $f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x, y)$, where the error term $E(x, y)$ satisfies $\lim_{(x,y)\to(x_0,y_0)} \frac{E(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0$

differentiable at

differentiable function | a function for which $f'(x)$ exists is a differentiable function

differentiable on

differential | the differential dx is an independent variable that can be assigned any nonzero real number; the differential dy is defined to be $dy = f'(x)\,dx$

differential calculus | the field of calculus concerned with the study of derivatives and their applications

differential equation | an equation involving a function $y = y(x)$ and one or more of its derivatives

differential form | given a differentiable function $y = f'(x)$, the equation $dy = f'(x)\,dx$ is the differential form of the derivative of y with respect to x

differentiation | the process of taking a derivative

direction angles | the angles formed by a nonzero vector and the coordinate axes

direction cosines | the cosines of the angles formed by a nonzero vector and the coordinate axes

direction field (slope field) | a mathematical object used to graphically represent solutions to a first-order differential equation; at each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point

direction vector | a vector parallel to a line that is used to describe the direction, or orientation, of the line in space

directional derivative | the derivative of a function in the direction of a given unit vector

directrix | a directrix (plural: directrices) is a line used to construct and define a conic section; a parabola has one directrix; ellipses and hyperbolas have two

discontinuity at a point | A function is discontinuous at a point or has a discontinuity at a point if it is not continuous at the point

discriminant | the value $4AC - B^2$, which is used to identify a conic when the equation contains a term involving xy, is called a discriminant

discriminant | the discriminant of the function $f(x, y)$ is given by the formula $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$

disk method | a special case of the slicing method used with solids of revolution when the slices are disks

divergence | the divergence of a vector field $\vec{\mathbf{F}} = \langle P, Q, R\rangle$, denoted $\vec{\nabla} \times \vec{\mathbf{F}}$, is $P_x + Q_y + R_z$; it measures the "outflowing-ness" of a vector field

divergence of a series | a series diverges if the sequence of partial sums for that series diverges

divergence test | if $\lim\limits_{n\to\infty} a_n \neq 0$, then the series $\sum\limits_{n=1}^{\infty} a_n$ diverges

divergent sequence | a sequence that is not convergent is divergent

domain | the set of inputs for a function

dot product or scalar product | $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = u_1v_1 + u_2v_2 + u_3v_3$ where $\vec{\mathbf{u}} = \langle u_1, u_2, u_3\rangle$ and $\vec{\mathbf{v}} = \langle v_1, v_2, v_3\rangle$

double integral | of the function $f(x, y)$ over the region R in the xy-plane is defined as the limit of a double Riemann sum,

$$\iint_R f(x, y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\,\Delta A.$$

double Riemann sum | of the function $f(x, y)$ over a rectangular region R is

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\,\Delta A,$$

where R is divided into smaller subrectangles R_{ij} and (x_{ij}^*, y_{ij}^*) is an arbitrary point in R_{ij}

doubling time | if a quantity grows exponentially, the doubling time is the amount of time it takes the quantity to double, and is given by $(\ln 2)/k$

eccentricity | the eccentricity is defined as the distance from any point on the conic section to its focus divided by the perpendicular distance from that point to the nearest directrix

ellipsoid | a three-dimensional surface described by an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$; all traces of this surface are ellipses

elliptic cone | a three-dimensional surface described by an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$; traces of this surface include ellipses and intersecting lines

elliptic paraboloid | a three-dimensional surface described by an equation of the form $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$; traces of this surface include ellipses and parabolas

end behavior | the behavior of a function as $x \to \infty$ and $x \to -\infty$

epsilon-delta definition of the limit | $\lim\limits_{x\to a} f(x) = L$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$

equilibrium solution | any solution to the differential equation of the form $y = c$, where c is a constant

equivalent vectors | vectors that have the same magnitude and the same direction

Euler's Method | a numerical technique used to approximate solutions to an initial-value problem

even function | a function is even if $f(-x) = f(x)$ for all x in the domain of f

explicit formula | a sequence may be defined by an explicit formula such that $a_n = f(n)$

exponent | the value x in the expression b^x

exponential decay | systems that exhibit exponential decay follow a model of the form $y = y_0e^{-kt}$

exponential growth | systems that exhibit exponential growth follow a model of the form $y = y_0e^{kt}$

extreme value theorem | if f is a continuous function over a finite, closed interval, then f has an absolute maximum and an absolute minimum

Fermat's theorem | if f has a local extremum at c, then c is a critical point of f

first derivative test | let f be a continuous function over an interval I containing a critical point c such that f is differentiable over I except possibly at c; if f' changes sign from positive to negative as x increases through c, then f has a local maximum at c; if f' changes sign from negative to positive as x increases through c, then f has a local minimum at c; if f' does not change sign as x increases through c, then f does not have a local extremum at c

flux | the rate of a fluid flowing across a curve in a vector field; the flux of vector field $\vec{\mathbf{F}}$ across plane curve C is line integral $\int_C \vec{\mathbf{F}} \cdot \frac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|}\, ds$

flux integral | another name for a surface integral of a vector field; the preferred term in physics and engineering

focal parameter | the focal parameter is the distance from a focus of a conic section to the nearest directrix

focus | a focus (plural: foci) is a point used to construct and define a conic section; a parabola has one focus; an ellipse and a hyperbola have two

formal definition of an infinite limit | $\lim\limits_{x\to a} f(x) = \infty$ if for every $M > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) > M$ $\lim\limits_{x\to a} f(x) = -\infty$ if for every $M > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) < -M$

Frenet frame of reference | (TNB frame) a frame of reference in three-dimensional space formed by the unit tangent vector, the unit normal vector, and the binormal vector

frustum | a portion of a cone; a frustum is constructed by cutting the cone with a plane parallel to the base

Fubini's theorem | if $f(x, y)$ is a function of two variables that is continuous over a rectangular region $R = \{(x, y) \in \mathbb{R}^2 \,|\, a \le x \le b, c \le y \le d\}$, then the double integral of f over the region equals an iterated integral,

$$\iint_R f(x,y)\, dA = \int_a^b \int_c^d f(x,y)\, dx\, dy = \int_c^d \int_a^b f(x,y)\, dx\, dy$$

function | a set of inputs, a set of outputs, and a rule for mapping each input to exactly one output

function of two variables | a function $z = f(x, y)$ that maps each ordered pair (x, y) in a subset D of R^2 to a unique real number z

Fundamental Theorem for Line Integrals | the value of line integral $\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}}$ depends only on the value of f at the endpoints of C : $\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a))$

fundamental theorem of calculus | (also, evaluation theorem) we can evaluate a definite integral by evaluating the antiderivative of the integrand at the endpoints of the interval and subtracting

fundamental theorem of calculus | uses a definite integral to define an antiderivative of a function

fundamental theorem of calculus | the theorem, central to the entire development of calculus, that establishes the relationship between differentiation and integration

general form | an equation of a conic section written as a general second-degree equation

general form of the equation of a plane | an equation in the form $ax + by + cz + d = 0$, where $\vec{\mathbf{n}} = \langle a, b, c\rangle$ is a normal vector of the plane, $P = (x_0, y_0, z_0)$ is a point on the plane, and $d = -ax_0 - by_0 - cz_0$

general solution (or family of solutions) | the entire set of solutions to a given differential equation

generalized chain rule | the chain rule extended to functions of more than one independent variable, in which each independent variable may depend on one or more other variables

geometric sequence | a sequence a_n in which the ratio a_{n+1}/a_n is the same for all positive integers n is called a geometric sequence

geometric series | a geometric series is a series that can be written in the form $\sum\limits_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \cdots$

gradient | the gradient of the function $f(x, y)$ is defined to be $\vec{\nabla} f(x, y) = (\partial f/\partial x)\,\hat{\mathbf{i}} + (\partial f/\partial y)\,\hat{\mathbf{j}}$, which can be generalized to a function of any number of independent variables

gradient field | a vector field $\vec{\mathbf{F}}$ for which there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$; in other words, a vector field that is the gradient of a function; such vector fields are also called *conservative*

graph of a function | the set of points (x, y) such that x is in the domain of f and $y = f(x)$

graph of a function of two variables | a set of ordered triples (x, y, z) that satisfies the equation $z = f(x, y)$ plotted in three-dimensional Cartesian space

Green's theorem | relates the integral over a connected region to an integral over the boundary of the region

grid curves | curves on a surface that are parallel to grid lines in a coordinate plane

growth rate | the constant $r > 0$ in the exponential growth function $P(t) = P_0 e^{rt}$

half-life | if a quantity decays exponentially, the half-life is the amount of time it takes the quantity to be reduced by half. It is given by $(\ln 2)/k$

harmonic series | the harmonic series takes the form $\sum\limits_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots$

heat flow | a vector field proportional to the negative temperature gradient in an object

helix | a three-dimensional curve in the shape of a spiral

higher-order derivative | a derivative of a derivative, from the second derivative to the nth derivative, is called a higher-order derivative

higher-order partial derivatives | second-order or higher partial derivatives, regardless of whether they are mixed partial derivatives

homogeneous linear equation | a second-order differential equation that can be written in the form $a_2(x)y''+a_1(x)y'+a_0(x)y=r(x)$, but $r(x)=0$ for every value of x

Hooke's law | this law states that the force required to compress (or elongate) a spring is proportional to the distance the spring has been compressed (or stretched) from equilibrium; in other words, $F=kx$, where k is a constant

horizontal asymptote | if $\lim\limits_{x\to\infty} f(x)=L$ or $\lim\limits_{x\to-\infty} f(x)=L$, then $y=L$ is a horizontal asymptote of f

horizontal line test | a function f is one-to-one if and only if every horizontal line intersects the graph of f, at most, once

hydrostatic pressure | the pressure exerted by water on a submerged object

hyperbolic functions | the functions denoted $\sinh$, $\cosh$, $\tanh$, csch, sech, and $\coth$, which involve certain combinations of e^x and e^{-x}

hyperboloid of one sheet | a three-dimensional surface described by an equation of the form $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$; traces of this surface include ellipses and hyperbolas

hyperboloid of two sheets | a three-dimensional surface described by an equation of the form $\frac{z^2}{c^2}-\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$; traces of this surface include ellipses and hyperbolas

implicit differentiation | is a technique for computing $\frac{dy}{dx}$ for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for $\frac{dy}{dx}$

improper double integral | a double integral over an unbounded region or of an unbounded function

improper integral | an integral over an infinite interval or an integral of a function containing an infinite discontinuity on the interval; an improper integral is defined in terms of a limit. The improper integral converges if this limit is a finite real number; otherwise, the improper integral diverges

increasing on the interval

indefinite integral | the most general antiderivative of $f(x)$ is the indefinite integral of f; we use the notation $\int f(x)\,dx$ to denote the indefinite integral of f

indefinite integral of a vector-valued function | a vector-valued function with a derivative that is equal to a given vector-valued function

independence of path | a vector field $\vec{\mathbf{F}}$ has path independence if $\int_{C_1}\vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}=\int_{C_2}\vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}$ for any curves C_1 and C_2 in the domain of $\vec{\mathbf{F}}$ with the same initial points and terminal points

independent variable | the input variable for a function

indeterminate forms | When evaluating a limit, the forms $\frac{0}{0}, \infty/\infty, 0\cdot\infty, \infty-\infty, 0^0, \infty^0$, and 1^∞ are considered indeterminate because further analysis is required to determine whether the limit exists and, if so, what its value is.

index variable | the subscript used to define the terms in a sequence is called the index

infinite discontinuity | An infinite discontinuity occurs at a point a if $\lim\limits_{x\to a^-} f(x)=\pm\infty$ or $\lim\limits_{x\to a^+} f(x)=\pm\infty$

infinite limit | A function has an infinite limit at a point a if it either increases or decreases without bound as it approaches a

infinite limit at infinity | a function that becomes arbitrarily large as x becomes large

infinite series | an infinite series is an expression of the form $a_1+a_2+a_3+\cdots=\sum\limits_{n=1}^{\infty}a_n$

inflection point | if f is continuous at c and f changes concavity at c, the point $(c,f(c))$ is an inflection point of f

initial point | the starting point of a vector

initial population | the population at time $t=0$

initial value problem | a problem that requires finding a function y that satisfies the differential equation $\frac{dy}{dx}=f(x)$ together with the initial condition $y(x_0)=y_0$

initial value(s) | a value or set of values that a solution of a differential equation satisfies for a fixed value of the independent variable

initial velocity | the velocity at time $t=0$

initial-value problem | a differential equation together with an initial value or values

instantaneous rate of change | the rate of change of a function at any point along the function a, also called $f'(a)$, or the derivative of the function at a

instantaneous velocity | The instantaneous velocity of an object with a position function that is given by $s(t)$ is the value that the average velocities on intervals of the form $[t,a]$ and $[a,t]$ approach as the values of t move closer to a, provided such a value exists

integrable function | a function is integrable if the limit defining the integral exists; in other words, if the limit of the Riemann sums as n goes to infinity exists

integral calculus | the study of integrals and their applications

integral test | for a series $\sum_{n=1}^{\infty} a_n$ with positive terms a_n, if there exists a continuous, decreasing function f such that $f(n) = a_n$ for all positive integers n, then

$$\sum_{n=1}^{\infty} a_n$$

and

$$\int_1^{\infty} f(x)\,dx$$

either both converge or both diverge

integrand | the function to the right of the integration symbol; the integrand includes the function being integrated

integrating factor | any function $f(x)$ that is multiplied on both sides of a differential equation to make the side involving the unknown function equal to the derivative of a product of two functions

integration by parts | a technique of integration that allows the exchange of one integral for another using the formula $\int u\,dv = uv - \int v\,du$

integration by substitution | a technique for integration that allows integration of functions that are the result of a chain-rule derivative

integration table | a table that lists integration formulas

interior point | a point P_0 of $\mathbb{R}$ is a boundary point if there is a δ disk centered around P_0 contained completely in $\mathbb{R}$

Intermediate Value Theorem | Let f be continuous over a closed bounded interval $[a, b]$ if z is any real number between $f(a)$ and $f(b)$, then there is a number c in [a, b] satisfying $f(c) = z$

intermediate variable | given a composition of functions (e.g., $f(x(t), y(t))$), the intermediate variables are the variables that are independent in the outer function but dependent on other variables as well; in the function $f(x(t), y(t))$, the variables x and y are examples of intermediate variables

interval of convergence | the set of real numbers x for which a power series converges

intuitive definition of the limit | If all values of the function $f(x)$ approach the real number L as the values of $x(\neq a)$ approach a, $f(x)$ approaches L

inverse function | for a function f, the inverse function f^{-1} satisfies $f^{-1}(y) = x$ if $f(x) = y$

inverse hyperbolic functions | the inverses of the hyperbolic functions where $\cosh$ and sech are restricted to the domain $[0, \infty)$;each of these functions can be expressed in terms of a composition of the natural logarithm function and an algebraic function

inverse trigonometric functions | the inverses of the trigonometric functions are defined on restricted domains where they are one-to-one functions

iterated integral | for a function $f(x, y)$ over the region R is a. $\int_a^b \int_c^d f(x, y)\,dx\,dy = \int_a^b \left[\int_c^d f(x, y)\,dy\right] dx$, b. $\int_c^d \int_a^b f(x, y)\,dx\,dy = \int_c^d \left[\int_a^b f(x, y)\,dx\right] dy$, where a, b, c, and d are any real numbers and $R = [a, b] \times [c, d]$

iterative process | process in which a list of numbers $x_0, x_1, x_2, x_3 \ldots$ is generated by starting with a number x_0 and defining $x_n = F(x_{n-1})$ for $n \geq 1$

Jacobian | the Jacobian $J(u, v)$ in two variables is a 2×2 determinant:

$$J(u, v) = \left|\begin{matrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{matrix}\right|;$$

the Jacobian $J(u, v, w)$ in three variables is a 3×3 determinant:

$$J(u, v, w) = \left|\begin{matrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial v} & \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{matrix}\right|$$

jump discontinuity | A jump discontinuity occurs at a point a if $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist, but $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$

Kepler's laws of planetary motion | three laws governing the motion of planets, asteroids, and comets in orbit around the Sun

Lagrange multiplier | the constant (or constants) used in the method of Lagrange multipliers; in the case of one constant, it is represented by the variable λ

lamina | a thin sheet of material; laminas are thin enough that, for mathematical purposes, they can be treated as if they are two-dimensional

left-endpoint approximation | an approximation of the area under a curve computed by using the left endpoint of each subinterval to calculate the height of the vertical sides of each rectangle

level curve of a function of two variables | the set of points satisfying the equation $f(x, y) = c$ for some real number c in the range of f

level surface of a function of three variables | the set of points satisfying the equation $f(x, y, z) = c$ for some real number c in the range of f

limaçon | the graph of the equation $r = a + b\sin\theta$ or $r = a + b\cos\theta$. If $a = b$ then the graph is a cardioid

limit | the process of letting x or t approach a in an expression; the limit of a function $f(x)$ as x approaches a is the value that $f(x)$ approaches as x approaches a

limit at infinity | a function that approaches a limit value L as x becomes large

limit comparison test | Suppose $a_n, b_n \geq 0$ for all $n \geq 1$. If $\lim_{n \to \infty} a_n/b_n \to L \neq 0$, then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both converge or both diverge; if $\lim_{n \to \infty} a_n/b_n \to 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. If $\lim_{n \to \infty} a_n/b_n \to \infty$, and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.

limit laws | the individual properties of limits; for each of the individual laws, let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a; assume that L and M are real numbers so that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$; let c be a constant

limit of a sequence | the real number LL to which a sequence converges is called the limit of the sequence

limit of a vector-valued function | a vector-valued function $\vec{\mathbf{r}}(t)$ has a limit $\vec{\mathbf{L}}$ as t approaches a if $\lim t \to a \left|\vec{\mathbf{r}}(t) - \vec{\mathbf{L}}\right| = 0$

limits of integration | these values appear near the top and bottom of the integral sign and define the interval over which the function should be integrated

line integral | the integral of a function along a curve in a plane or in space

linear | description of a first-order differential equation that can be written in the form $a(x)y' + b(x)y = c(x)$

linear approximation | the linear function $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of f at $x = a$

linear approximation | given a function $f(x, y)$ and a tangent plane to the function at a point (x_0, y_0), we can approximate $f(x, y)$ for points near (x_0, y_0) using the tangent plane formula

linear function | a function that can be written in the form $f(x) = mx + b$

linearly dependent | a set of functions $f_1(x), f_2(x), \ldots, f_n(x)$ for whichthere are constants $c_1, c_2, \ldots c_n$, not all zero, such that $c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$ for all \ (x\) in the interval of interest

linearly independent | a set of functions $f_1(x), f_2(x), \ldots, f_n(x)$ for which there are no constants $c_1, c_2, \ldots c_n$, such that $c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$ for all \(x\) in the interval of interest

local extremum | if f has a local maximum or local minimum at c, we say f has a local extremum at c

local maximum | if there exists an interval I such that $f(c) \geq f(x)$ for all $x \in I$, we say f has a local maximum at c

local minimum | if there exists an interval I such that $f(c) \leq f(x)$ for all $x \in I$, we say f has a local minimum at c

logarithmic differentiation | is a technique that allows us to differentiate a function by first taking the natural logarithm of both sides of an equation, applying properties of logarithms to simplify the equation, and differentiating implicitly

logarithmic function | a function of the form $f(x) = \log_b(x)$ for some base $b > 0$, $b \neq 1$ such that $y = \log_b(x)$ if and only if $b^y = x$

logistic differential equation | a differential equation that incorporates the carrying capacity K and growth rate rr into a population model

lower sum | a sum obtained by using the minimum value of $f(x)$ on each subinterval

L'Hôpital's rule | If f and g are differentiable functions over an interval a, except possibly at a, and $\lim_{x \to a} f(x) = 0 = \lim_{x \to a} g(x)$ or $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ are infinite, then $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$, assuming the limit on the right exists or is ∞ or $-\infty$.

Maclaurin polynomial | a Taylor polynomial centered at 0; the nth-degree Taylor polynomial for f at 0 is the nth-degree Maclaurin polynomial for f

Maclaurin series | a Taylor series for a function f at $x = 0$ is known as a Maclaurin series for f

magnitude | the length of a vector

major axis | the major axis of a conic section passes through the vertex in the case of a parabola or through the two vertices in the case of an ellipse or hyperbola; it is also an axis of symmetry of the conic; also called the transverse axis

marginal cost | is the derivative of the cost function, or the approximate cost of producing one more item

marginal profit | is the derivative of the profit function, or the approximate profit obtained by producing and selling one more item

marginal revenue | is the derivative of the revenue function, or the approximate revenue obtained by selling one more item

mass flux | the rate of mass flow of a fluid per unit area, measured in mass per unit time per unit area

mathematical model | A method of simulating real-life situations with mathematical equations

mean value theorem | if f is continuous over $[a, b]$ and differentiable over (a, b), then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

mean value theorem for integrals | guarantees that a point c exists such that $f(c)$ is equal to the average value of the function

method of cylindrical shells | a method of calculating the volume of a solid of revolution by dividing the solid into nested cylindrical shells; this method is different from the methods of disks or washers in that we integrate with respect to the opposite variable

method of Lagrange multipliers | a method of solving an optimization problem subject to one or more constraints

method of undetermined coefficients | a method that involves making a guess about the form of the particular solution, then solving for the coefficients in the guess

method of variation of parameters | a method that involves looking for particular solutions in the form $y_p(x) = u(x)y_1(x) + v(x)y_2(x)$, where y_1 and y_2 are linearly independent solutions to the complementary equations, and then solving a system of equations to find $u(x)$ and $v(x)$

midpoint rule | a rule that uses a Riemann sum of the form $M_n = \sum_{i=1}^{n} f(m_i)\Delta x$, where m_i is the midpoint of the ith subinterval to approximate $\int_a^b f(x)\,dx$

minor axis | the minor axis is perpendicular to the major axis and intersects the major axis at the center of the conic, or at the vertex in the case of the parabola; also called the conjugate axis

mixed partial derivatives | second-order or higher partial derivatives, in which at least two of the differentiations are with respect to different variables

moment | if n masses are arranged on a number line, the moment of the system with respect to the origin is given by $M = \sum_{i=1}^{n} m_i x_i$; if, instead, we consider a region in the plane, bounded above by a function $f(x)$ over an interval $[a, b]$, then the moments of the region with respect to the x- and y-axes are given by $M_x = \rho \int_a^b \frac{[f(x)]^2}{2}\,dx$ and $M_y = \rho \int_a^b x f(x)\,dx$, respectively

monotone sequence | an increasing or decreasing sequence

multivariable calculus | the study of the calculus of functions of two or more variables

nappe | a nappe is one half of a double cone

natural exponential function | the function $f(x) = e^x$

natural logarithm | the function $\ln x = \log_e x$

net change theorem | if we know the rate of change of a quantity, the net change theorem says the future quantity is equal to the initial quantity plus the integral of the rate of change of the quantity

net signed area | the area between a function and the x-axis such that the area below the x-axis is subtracted from the area above the x-axis; the result is the same as the definite integral of the function

Newton's method | method for approximating roots of $f(x) = 0$; using an initial guess x_0; each subsequent approximation is defined by the equation $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$

nonelementary integral | an integral for which the antiderivative of the integrand cannot be expressed as an elementary function

nonhomogeneous linear equation | a second-order differential equation that can be written in the form $a_2(x)y'' + a_1(x)y' + a_0(x)y = r(x)$, but $r(x) \neq 0$ for some value of x

normal component of acceleration | the coefficient of the unit normal vector $\vec{\mathrm{N}}$ when the acceleration vector is written as a linear combination of $\vec{\mathrm{T}}$ and $\vec{\mathrm{N}}$

normal plane | a plane that is perpendicular to a curve at any point on the curve

normal vector | a vector perpendicular to a plane

normalization | using scalar multiplication to find a unit vector with a given direction

number e | as m gets larger, the quantity $(1 + (1/m)^m$ gets closer to some real number; we define that real number to be e; the value of e is approximately 2.718282

numerical integration | the variety of numerical methods used to estimate the value of a definite integral, including the midpoint rule, trapezoidal rule, and Simpson's rule

objective function | the function that is to be maximized or minimized in an optimization problem

oblique asymptote | the line $y = mx + b$ if $f(x)$ approaches it as $x \to \infty$ or$x \to -\infty$

octants | the eight regions of space created by the coordinate planes

odd function | a function is odd if $f(-x) = -f(x)$ for all x in the domain of f

one-sided limit | A one-sided limit of a function is a limit taken from either the left or the right

one-to-one function | a function f is one-to-one if $f(x_1) \neq f(x_2)$ if $x_1 \neq x_2$

one-to-one transformation | a transformation $T : G \to R$ defined as $T(u, v) = (x, y)$ is said to be one-to-one if no two points map to the same image point

open set | a set S that contains none of its boundary points

optimization problem | calculation of a maximum or minimum value of a function of several variables, often using Lagrange multipliers

optimization problems | problems that are solved by finding the maximum or minimum value of a function

order of a differential equation | the highest order of any derivative of the unknown function that appears in the equation

orientation | the direction that a point moves on a graph as the parameter increases

orientation of a curve | the orientation of a curve C is a specified direction of C

orientation of a surface | if a surface has an "inner" side and an "outer" side, then an orientation is a choice of the inner or the outer side; the surface could also have "upward" and "downward" orientations

orthogonal vectors | vectors that form a right angle when placed in standard position

osculating circle | a circle that is tangent to a curve C at a point P and that shares the same curvature

osculating plane | the plane determined by the unit tangent and the unit normal vector

***p*-series** | a series of the form $\sum_{n=1}^{\infty} 1/n^p$

parallelogram method | a method for finding the sum of two vectors; position the vectors so they share the same initial point; the vectors then form two adjacent sides of a parallelogram; the sum of the vectors is the diagonal of that parallelogram

parameter | an independent variable that both x and y depend on in a parametric curve; usually represented by the variable t

parameter domain (parameter space) | the region of the uv-plane over which the parameters u and v vary for parameterization $\vec{\mathbf{r}}(u, v) = \langle x(u, v),\ y(u, v),\ z(u, v)\rangle$

parameterization of a curve | rewriting the equation of a curve defined by a function $y = f(x)$ as parametric equations

parameterized surface (parametric surface) | a surface given by a description of the form $\vec{\mathbf{r}}(u, v) = \langle x(u, v),\ y(u, v),\ z(u, v)\rangle$, where the parameters u and v vary over a parameter domain in the uv-plane

parametric curve | the graph of the parametric equations $x(t)$ and $y(t)$ over an interval $a \leq t \leq b$ combined with the equations

parametric equations | the equations $x = x(t)$ and $y = y(t)$ that define a parametric curve

parametric equations of a line | the set of equations $x = x_0 + ta, y = y_0 + tb$, and $z = z_0 + tc$ describing the line with direction vector $v = \langle a, b, c\rangle$ passing through point (x_0, y_0, z_0)

partial derivative | a derivative of a function of more than one independent variable in which all the variables but one are held constant

partial differential equation | an equation that involves an unknown function of more than one independent variable and one or more of its partial derivatives

partial fraction decomposition | a technique used to break down a rational function into the sum of simple rational functions

partial sum | the kth partial sum of the infinite series $\sum_{n=1}^{\infty} a_n$ is the finite sum $S_k = \sum_{n=1}^{k} a_n = a_1 + a_2 + a_3 + \cdots + a_k$

particular solution | member of a family of solutions to a differential equation that satisfies a particular initial condition

particular solution | a solution $y_p(x)$ of a differential equation that contains no arbitrary constants

partition | a set of points that divides an interval into subintervals

percentage error | the relative error expressed as a percentage

periodic function | a function is periodic if it has a repeating pattern as the values of x move from left to right

phase line | a visual representation of the behavior of solutions to an autonomous differential equation subject to various initial conditions

piecewise smooth curve | an oriented curve that is not smooth, but can be written as the union of finitely many smooth curves

piecewise-defined function | a function that is defined differently on different parts of its domain

planar transformation | a function T that transforms a region G in one plane into a region R in another plane by a change of variables

plane curve | the set of ordered pairs $(f(t), g(t))$ together with their defining parametric equations $x = f(t)$ and $y = g(t)$

point-slope equation | equation of a linear function indicating its slope and a point on the graph of the function

polar axis | the horizontal axis in the polar coordinate system corresponding to $r \geq 0$

polar coordinate system | a system for locating points in the plane. The coordinates are r, the radial coordinate, and θ, the angular coordinate

polar equation | an equation or function relating the radial coordinate to the angular coordinate in the polar coordinate system

polar rectangle | the region enclosed between the circles $r = a$ and $r = b$ and the angles $\theta = \alpha$ and $\theta = \beta$; it is described as $R = \{(r, \theta) \mid a \leq r \leq b,\ \alpha \leq \theta \leq \beta\}$

pole | the central point of the polar coordinate system, equivalent to the origin of a Cartesian system

polynomial function | a function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$

population growth rate | is the derivative of the population with respect to time

potential function | a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$

power function | a function of the form $f(x) = x^n$ for any positive integer $n \geq 1$

power law for limits | the limit law

$$\begin{aligned} &\lim_{x \to a} (f(x))^n \\ &= (\lim_{x \to a} f(x))^n \\ &= L^n \end{aligned}$$

for every positive integer n

power reduction formula | a rule that allows an integral of a power of a trigonometric function to be exchanged for an integral involving a lower power

power rule | the derivative of a power function is a function in which the power on x becomes the coefficient of the term and the power on x in the derivative decreases by 1: If n is an integer, then $\frac{d}{dx}(x^n) = nx^{n-1}$

power series | a series of the form $\sum_{n=0}^{\infty} c_n x^n$ is a power series centered at $x = 0$; a series of the form $\sum_{n=0}^{\infty} c_n (x - a)^n$ is a power series centered at $x = a$

principal unit normal vector | a vector orthogonal to the unit tangent vector, given by the formula $\frac{\vec{\mathbf{T}}'(t)}{\|\vec{\mathbf{T}}'(t)\|}$

principal unit tangent vector | a unit vector tangent to a curve C

product law for limits | the limit law

$$\begin{aligned} &\lim_{x \to a} (f(x) \cdot g(x)) \\ &= \lim_{x \to a} f(x) \\ &\cdot \lim_{x \to a} g(x) \\ &= L \cdot M \end{aligned}$$

product rule | the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function: $\frac{d}{dx}\big(f(x)g(x)\big) = f'(x)g(x) + g'(x)f(x)$

projectile motion | motion of an object with an initial velocity but no force acting on it other than gravity

propagated error | the error that results in a calculated quantity $f(x)$ resulting from a measurement error dx

quadratic function | a polynomial of degree 2; that is, a function of the form $f(x) = ax^2 + bx + c$ where $a \neq 0$

quadric surfaces | surfaces in three dimensions having the property that the traces of the surface are conic sections (ellipses, hyperbolas, and parabolas)

quotient law for limits | the limit law $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{L}{M}$ for M≠0

quotient rule | the derivative of the quotient of two functions is the derivative of the first function times the second function minus the derivative of the second function times the first function, all divided by the square of the second function: $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - g'(x)f(x)}{\big(g(x)\big)^2}$

radial coordinate | r the coordinate in the polar coordinate system that measures the distance from a point in the plane to the pole

radial field | a vector field in which all vectors either point directly toward or directly away from the origin; the magnitude of any vector depends only on its distance from the origin

radians | for a circular arc of length s on a circle of radius 1, the radian measure of the associated angle θ is s

radius of convergence | if there exists a real number $R > 0$ such that a power series centered at $x = a$ converges for $|x - a| < R$ and diverges for $|x - a| > R$, then R is the radius of convergence; if the power series only converges at $x = a$, the radius of convergence is $R = 0$; if the power series converges for all real numbers x, the radius of convergence is $R = \infty$

radius of curvature | the reciprocal of the curvature

radius of gyration | the distance from an object's center of mass to its axis of rotation

range | the set of outputs for a function

ratio test | for a series $\sum_{n=1}^{\infty} a_n$ with nonzero terms, let $\rho = \lim_{n\to\infty} |a_{n+1}/a_n|$; if $0 \le \rho < 1$, the series converges absolutely; if $\rho > 1$, the series diverges; if $\rho = 1$, the test is inconclusive

rational function | a function of the form $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomials

recurrence relation | a recurrence relation is a relationship in which a term a_n in a sequence is defined in terms of earlier terms in the sequence

region | an open, connected, nonempty subset of $\mathbb{R}^2$

regular parameterization | parameterization $\vec{r}(u, v) = \langle x(u, v),\ y(u, v),\ z(u, v)\rangle$ such that $r_u \times r_v$ is not zero for point (u, v) in the parameter domain

regular partition | a partition in which the subintervals all have the same width

related rates | are rates of change associated with two or more related quantities that are changing over time

relative error | given an absolute error Δq for a particular quantity, $\frac{\Delta q}{q}$ is the relative error.

relative error | error as a percentage of the actual value, given by

$$\begin{matrix}\text{relative error} \\ = \left|\frac{A - B}{A}\right| \\ \cdot\, 100\%\end{matrix}$$

remainder estimate | for a series $\sum_{n=}^{\infty} 1a_n$ with positive terms a_n and a continuous, decreasing function f such that $f(n) = a_n$ for all positive integers n, the remainder $R_N = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{N} a_n$ satisfies the following estimate:

$$\begin{matrix}\int_{N+1}^{\infty} f(x)\,dx \\ < R_N \\ < \int_{N}^{\infty} f(x)\,dx\end{matrix}$$

removable discontinuity | A removable discontinuity occurs at a point a if $f(x)$ is discontinuous at a, but $\lim_{x\to a} f(x)$ exists

reparameterization | an alternative parameterization of a given vector-valued function

restricted domain | a subset of the domain of a function f

riemann sum | an estimate of the area under the curve of the form $A \approx \sum_{i=1}^{n} f(x_i^*)\Delta x$

right-endpoint approximation | the right-endpoint approximation is an approximation of the area of the rectangles under a curve using the right endpoint of each subinterval to construct the vertical sides of each rectangle

right-hand rule | a common way to define the orientation of the three-dimensional coordinate system; when the right hand is curved around the z-axis in such a way that the fingers curl from the positive x-axis to the positive y-axis, the thumb points in the direction of the positive z-axis

***RLC* series circuit** | a complete electrical path consisting of a resistor, an inductor, and a capacitor; a second-order, constant-coefficient differential equation can be used to model the charge on the capacitor in an *RLC* series circuit

rolle's theorem | if f is continuous over $[a, b]$ and differentiable over (a, b), and if $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$

root function | a function of the form $f(x) = x^{1/n}$ for any integer $n \ge 2$

root law for limits | the limit law $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)} = \sqrt[n]{L}$ for all L if n is odd and for $L \ge 0$ if n is even

root test | for a series $\sum_{n=1}^{\infty} a_n$, let $\rho = \lim_{n\to\infty} \sqrt[n]{|a_n|}$; if $0 \le \rho < 1$, the series converges absolutely; if $\rho > 1$, the series diverges; if $\rho = 1$, the test is inconclusive

rose | graph of the polar equation $r = a\cos 2\theta$ or $r = a\sin 2\theta$for a positive constant a

rotational field | a vector field in which the vector at point (x, y) is tangent to a circle with radius $r = \sqrt{x^2 + y^2}$; in a rotational field, all vectors flow either clockwise or counterclockwise, and the magnitude of a vector depends only on its distance from the origin

rulings | parallel lines that make up a cylindrical surface

saddle point | given the function $z = f(x, y)$, the point $(x_0, y_0, f(x_0, y_0))$ is a saddle point if both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$, but f does not have a local extremum at (x_0, y_0)

scalar | a real number

scalar equation of a plane | the equation $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$ used to describe a plane containing point $P=(x_0,y_0,z_0)$ with normal vector $n=\langle a,b,c\rangle$ or its alternate form $ax+by+cz+d=0$, where $d=-ax_0-by_0-cz_0$

scalar line integral | the scalar line integral of a function f along a curve C with respect to arc length is the integral $\int_C f\,ds$, it is the integral of a scalar function f along a curve in a plane or in space; such an integral is defined in terms of a Riemann sum, as is a single-variable integral

scalar multiplication | a vector operation that defines the product of a scalar and a vector

scalar projection | the magnitude of the vector projection of a vector

secant | A secant line to a function $f(x)$ at a is a line through the point $(a,f(a))$ and another point on the function; the slope of the secant line is given by $m_{sec}=\dfrac{f(x)-f(a)}{x-a}$

second derivative test | suppose $f'(c)=0$ and f'' is continuous over an interval containing c; if $f''(c)>0$, then f has a local minimum at c; if $f''(c)<0$, then f has a local maximum at c; if $f''(c)=0$, then the test is inconclusive

separable differential equation | any equation that can be written in the form $y'=f(x)g(y)$

separation of variables | a method used to solve a separable differential equation

sequence | an ordered list of numbers of the form $a_1,a_2,a_3,\ldots$ is a sequence

sigma notation | (also, summation notation) the Greek letter sigma (Σ) indicates addition of the values; the values of the index above and below the sigma indicate where to begin the summation and where to end it

simple curve | a curve that does not cross itself

simple harmonic motion | motion described by the equation $x(t)=c_1\cos(\omega t)+c_2\sin(\omega t)$, as exhibited by an undamped spring-mass system in which the mass continues to oscillate indefinitely

simply connected region | a region that is connected and has the property that any closed curve that lies entirely inside the region encompasses points that are entirely inside the region

Simpson's rule | a rule that approximates $\int_a^b f(x)\,dx$ using the area under a piecewise quadratic function. The approximation S_n to $\int_a^b f(x)\,dx$ is given by

$$\begin{aligned}S_n=\frac{\Delta x}{3}\big(&f(x_0)\\&+4f(x_1)\\&+2f(x_2)\\&+4f(x_3)\\&+2f(x_4)+\cdots\\&+2f(x_{n-2})\\&+4f(x_{n-1})\\&+f(x_n)\big).\end{aligned}$$

skew lines | two lines that are not parallel but do not intersect

slicing method | a method of calculating the volume of a solid that involves cutting the solid into pieces, estimating the volume of each piece, then adding these estimates to arrive at an estimate of the total volume; as the number of slices goes to infinity, this estimate becomes an integral that gives the exact value of the volume

slope | the change in y for each unit change in x

slope-intercept form | equation of a linear function indicating its slope and y-intercept

smooth | curves where the vector-valued function $\vec{\mathbf{r}}(t)$ is differentiable with a non-zero derivative

solid of revolution | a solid generated by revolving a region in a plane around a line in that plane

solution curve | a curve graphed in a direction field that corresponds to the solution to the initial-value problem passing through a given point in the direction field

solution to a differential equation | a function $y=f(x)$ that satisfies a given differential equation

space curve | the set of ordered triples $(f(t),g(t),h(t))$ together with their defining parametric equations $x=f(t)$, $y=g(t)$ and $z=h(t)$

space-filling curve | a curve that completely occupies a two-dimensional subset of the real plane

speed | is the absolute value of velocity, that is, $|v(t)|$ is the speed of an object at time t whose velocity is given by $v(t)$

sphere | the set of all points equidistant from a given point known as the *center*

spherical coordinate system | a way to describe a location in space with an ordered triple (ρ,θ,φ), where ρ is the distance between P and the origin $(\rho\neq 0)$, θ is the same angle used to describe the location in cylindrical coordinates, and φ is the angle formed by the positive z-axis and line segment $\overline{OP}$, where O is the origin and $0\leq\varphi\leq\pi$

squeeze theorem | states that if $f(x)\leq g(x)\leq h(x)$ for all $x\neq a$ over an open interval containing a and $\lim_{x\to a}f(x)=L=\lim_{x\to a}h(x)$ where L is a real number, then $\lim_{x\to a}g(x)=L$

standard equation of a sphere | $(x-a)^2+(y-b)^2+(z-c)^2=r^2$ describes a sphere with center (a,b,c) and radius r

standard form | the form of a first-order linear differential equation obtained by writing the differential equation in the form $y'+p(x)y=q(x)$

standard form | an equation of a conic section showing its properties, such as location of the vertex or lengths of major and minor axes

standard unit vectors | unit vectors along the coordinate axes: $\hat{\mathbf{i}}=\langle 1,0\rangle$, $\hat{\mathbf{j}}=\langle 0,1\rangle$

standard-position vector | a vector with initial point $(0,0)$

steady-state solution | a solution to a nonhomogeneous differential equation related to the forcing function; in the long term, the solution approaches the steady-state solution

step size | the increment hh that is added to the xx value at each step in Euler's Method

Stokes' theorem | relates the flux integral over a surface S to a line integral around the boundary C of the surface S

stream function | if $\vec{\mathbf{F}} = \langle P, Q \rangle$ is a source-free vector field, then stream function g is a function such that $P = g_y$ and $Q = -g_x$

sum law for limits | The limit law $\lim_{x \to a}(f(x) + g(x)) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = L + M$

sum rule | the derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g: $\frac{d}{dx}\big(f(x) + g(x)\big) = f'(x) + g'(x)$

surface | the graph of a function of two variables, $z = f(x, y)$

surface area | the surface area of a solid is the total area of the outer layer of the object; for objects such as cubes or bricks, the surface area of the object is the sum of the areas of all of its faces

surface area | the area of surface S given by the surface integral

$$\iint_S dS$$

surface independent | flux integrals of curl vector fields are surface independent if their evaluation does not depend on the surface but only on the boundary of the surface

surface integral | an integral of a function over a surface

surface integral of a scalar-valued function | a surface integral in which the integrand is a scalar function

surface integral of a vector field | a surface integral in which the integrand is a vector field

symmetric equations of a line | the equations $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$ describing the line with direction vector $v = \langle a, b, c \rangle$ passing through point (x_0, y_0, z_0)

symmetry about the

symmetry about the origin | the graph of a function f is symmetric about the origin if $(-x, -y)$ is on the graph of f whenever (x, y) is on the graph

symmetry principle | the symmetry principle states that if a region R is symmetric about a line I, then the centroid of R lies on I

table of values | a table containing a list of inputs and their corresponding outputs

tangent | A tangent line to the graph of a function at a point $(a, f(a))$ is the line that secant lines through $(a, f(a))$ approach as they are taken through points on the function with x-values that approach a; the slope of the tangent line to a graph at a measures the rate of change of the function at a

tangent line approximation (linearization) | since the linear approximation of f at $x = a$ is defined using the equation of the tangent line, the linear approximation of f at $x = a$ is also known as the tangent line approximation to f at $x = a$

tangent plane | given a function $f(x, y)$ that is differentiable at a point (x_0, y_0), the equation of the tangent plane to the surface $z = f(x, y)$ is given by $z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

tangent vector | to $\vec{\mathbf{r}}(t)$ at $t = t_0$ any vector $\vec{\mathbf{v}}$ such that, when the tail of the vector is placed at point $\vec{\mathbf{r}}(t_0)$ on the graph, vector $\vec{\mathbf{v}}$ is tangent to curve C

tangential component of acceleration | the coefficient of the unit tangent vector $\vec{\mathbf{T}}$ when the acceleration vector is written as a linear combination of $\vec{\mathbf{T}}$ and $\vec{\mathbf{N}}$

Taylor polynomials | the nth-degree Taylor polynomial for f at $x = a$ is $p_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$

Taylor series | a power series at a that converges to a function f on some open interval containing a.

Taylor's theorem with remainder | for a function f and the nth-degree Taylor polynomial for f at $x = a$, the remainder $R_n(x) = f(x) - p_n(x)$ satisfies $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1}$ for some$c$ between x and a; if there exists an interval I containing a and a real number M such that $|\, f^{(n+1)}(x) \,| \leq M$ for all x in I, then $|R_n(x)| \leq \frac{M}{(n+1)!}|x - a|^{n+1}$

telescoping series | a telescoping series is one in which most of the terms cancel in each of the partial sums

term | the number a_n in the sequence a_n is called the nth term of the sequence

term-by-term differentiation of a power series | a technique for evaluating the derivative of a power series $\sum_{n=0}^{\infty} c_n(x - a)^n$ by evaluating the derivative of each term separately to create the new power series $\sum_{n=1}^{\infty} n c_n(x - a)^{n-1}$

term-by-term integration of a power series | a technique for integrating a power series $\sum_{n=0}^{\infty} c_n(x - a)^n$ by integrating each term separately to create the new power series $C + \sum_{n=0}^{\infty} c_n \frac{(x - a)^{n+1}}{n + 1}$

terminal point | the endpoint of a vector

theorem of Pappus for volume | this theorem states that the volume of a solid of revolution formed by revolving a region around an external axis is equal to the area of the region multiplied by the distance traveled by the centroid of the region

three-dimensional rectangular coordinate system | a coordinate system defined by three lines that intersect at right angles; every point in space is described by an ordered triple (x, y, z) that plots its location relative to the defining axes

threshold population | the minimum population that is necessary for a species to survive

total area | total area between a function and the x-axis is calculated by adding the area above the x-axis and the area below the x-axis; the result is the same as the definite integral of the absolute value of the function

total differential | the total differential of the function $f(x, y)$ at (x_0, y_0) is given by the formula $dz = f_x(x_0, y_0)dx + fy(x_0, y_0)dy$

trace | the intersection of a three-dimensional surface with a coordinate plane

transcendental function | a function that cannot be expressed by a combination of basic arithmetic operations

transformation | a function that transforms a region GG in one plane into a region RR in another plane by a change of variables

transformation of a function | a shift, scaling, or reflection of a function

trapezoidal rule | a rule that approximates $\int_a^b f(x)\,dx$ using the area of trapezoids. The approximation T_n to $\int_a^b f(x)\,dx$ is given by

$$\begin{gathered} T_n = \frac{\Delta x}{2} \\ \big(f(x_0) \\ + 2\,f(x_1) \\ + 2\,f(x_2) + \cdots \\ + 2\,f(x_{n-1}) \\ + f(x_n)\big). \end{gathered}$$

tree diagram | illustrates and derives formulas for the generalized chain rule, in which each independent variable is accounted for

triangle inequality | If a and b are any real numbers, then $|a + b| \leq |a| + |b|$

triangle inequality | the length of any side of a triangle is less than the sum of the lengths of the other two sides

triangle method | a method for finding the sum of two vectors; position the vectors so the terminal point of one vector is the initial point of the other; these vectors then form two sides of a triangle; the sum of the vectors is the vector that forms the third side; the initial point of the sum is the initial point of the first vector; the terminal point of the sum is the terminal point of the second vector

trigonometric functions | functions of an angle defined as ratios of the lengths of the sides of a right triangle

trigonometric identity | an equation involving trigonometric functions that is true for all angles θ for which the functions in the equation are defined

trigonometric integral | an integral involving powers and products of trigonometric functions

trigonometric substitution | an integration technique that converts an algebraic integral containing expressions of the form $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$ into a trigonometric integral

triple integral | the triple integral of a continuous function $f(x, y, z)$ over a rectangular solid box B is the limit of a Riemann sum for a function of three variables, if this limit exists

triple integral in cylindrical coordinates | the limit of a triple Riemann sum, provided the following limit exists:

$$\begin{gathered} lim_{l,m,n\to\infty} \sum_{i=1}^{l} \\ \sum_{j=1}^{m} \\ \sum_{k=1}^{n} f(r^*_{ijk}, \theta^*_{ijk}, \\ s^*_{ijk}) r^*_{ijk} \Delta r \\ \Delta\theta \Delta z \end{gathered}$$

triple integral in spherical coordinates | the limit of a triple Riemann sum, provided the following limit exists:

$$\begin{gathered} lim_{l,m,n\to\infty} \sum_{i=1}^{l} \\ \sum_{j=1}^{m} \\ \sum_{k=1}^{n} f(\rho^*_{ijk}, \theta^*_{ijk}, \\ \varphi^*_{ijk}) (\rho^*_{ijk})^2 \\ \sin\varphi \Delta\rho \Delta\theta \Delta\varphi \end{gathered}$$

Type I | a region D in the xy- plane is Type I if it lies between two vertical lines and the graphs of two continuous functions $g_1(x)$ and $g_2(x)$

Type II | a region D in the xy-plane is Type II if it lies between two horizontal lines and the graphs of two continuous functions $h_1(y)$ and $h_2(h)$

unbounded sequence | a sequence that is not bounded is called unbounded

unit vector | a vector with magnitude 1

unit vector field | a vector field in which the magnitude of every vector is 1

upper sum | a sum obtained by using the maximum value of $f(x)$ on each subinterval

variable of integration | indicates which variable you are integrating with respect to; if it is x, then the function in the integrand is followed by dx

vector | a mathematical object that has both magnitude and direction

vector addition | a vector operation that defines the sum of two vectors

vector difference | the vector difference $\vec{\mathrm{v}} - \vec{\mathrm{w}}$ is defined as $\vec{\mathrm{v}} + (-\vec{\mathrm{w}}) = \vec{\mathrm{v}} + (-1)\vec{\mathrm{w}}$

vector equation of a line | the equation $\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + t\vec{\mathbf{v}}$ used to describe a line with direction vector $\vec{\mathbf{v}} = \langle a, b, c \rangle$ passing through point $P = (x_0, y_0, z_0)$, where $\vec{\mathbf{r}}_0 = \langle x_0, y_0, z_0 \rangle$, is the position vector of point P

vector equation of a plane | the equation $\vec{\mathbf{n}} \cdot \overrightarrow{PQ} = 0$, where P is a given point in the plane, Q is any point in the plane, and $\vec{\mathbf{n}}$ is a normal vector of the plane

vector field | measured in $\mathbb{R}^2$, an assignment of a vector $\vec{\mathbf{F}}(x, y)$ to each point (x, y) of a subset D of $\mathbb{R}^2$; in $\mathbb{R}^3$, an assignment of a vector $\vec{\mathbf{F}}(x, y, z)$ to each point (x, y, z) of a subset D of $\mathbb{R}^3$

vector line integral | the vector line integral of vector field $\vec{\mathbf{F}}$ along curve C is the integral of the dot product of $\vec{\mathbf{F}}$ with unit tangent vector $\vec{\mathbf{T}}$ of C with respect to arc length, $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$; such an integral is defined in terms of a Riemann sum, similar to a single-variable integral

vector parameterization | any representation of a plane or space curve using a vector-valued function

vector projection | the component of a vector that follows a given direction

vector sum | the sum of two vectors, $\vec{\mathbf{v}}$ and $\vec{\mathbf{w}}$, can be constructed graphically by placing the initial point of $\vec{\mathbf{w}}$ at the terminal point of $\vec{\mathbf{v}}$; then the vector sum $\vec{\mathbf{v}} + \vec{\mathbf{w}}$ is the vector with an initial point that coincides with the initial point of $\vec{\mathbf{v}}$, and with a terminal point that coincides with the terminal point of $\vec{\mathbf{w}}$

vector-valued function | a function of the form $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}}$ or $\vec{\mathbf{r}}(t) = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$,where the component functions f, g, and h are real-valued functions of the parameter t.

velocity vector | the derivative of the position vector

vertex | a vertex is an extreme point on a conic section; a parabola has one vertex at its turning point. An ellipse has two vertices, one at each end of the major axis; a hyperbola has two vertices, one at the turning point of each branch

vertical asymptote | A function has a vertical asymptote at $x = a$ if the limit as x approaches a from the right or left is infinite

vertical line test | given the graph of a function, every vertical line intersects the graph, at most, once

vertical trace | the set of ordered triples (c, y, z) that solves the equation $f(c, y) = z$ for a given constant $x = c$ or the set of ordered triples (x, d, z) that solves the equation $f(x, d) = z$ for a given constant $y = d$

washer method | a special case of the slicing method used with solids of revolution when the slices are washers

work | the amount of energy it takes to move an object; in physics, when a force is constant, work is expressed as the product of force and distance

work done by a force | work is generally thought of as the amount of energy it takes to move an object; if we represent an applied force by a vector $\vec{\mathbf{F}}$ and the displacement of an object by a vector $\vec{\mathbf{s}}$, then the work done by the force is the dot product of $\vec{\mathbf{F}}$ and $\vec{\mathbf{s}}$.

zero vector | the vector with both initial point and terminal point $(0, 0)$

zeros of a function | when a real number x is a zero of a function f, $f(x) = 0$

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    - 2.4: Continuity - *CC BY-NC-SA 4.0*
      - 2.4E: Exercises for Section 2.4 - *CC BY-NC-SA 4.0*
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      - 3.3E: Exercises for Section 3.3 - *CC BY-NC-SA 4.0*
    - 3.4: Derivatives as Rates of Change - *CC BY-NC-SA 4.0*
      - 3.4E: Exercises for Section 3.4 - *CC BY-NC-SA 4.0*
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      - 3.5E: Exercises for Section 3.5 - *CC BY-NC-SA 4.0*
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- 5.6: Integrals Involving Exponential and Logarithmic Functions - *CC BY-NC-SA 4.0*
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  - 10.2E: Exercises for Section 10.2 - *CC BY-NC-SA 4.0*
- 10.3: Taylor and Maclaurin Series - *CC BY-NC-SA 4.0*
  - 10.3E: Exercises for Section 10.3 - *CC BY-NC-SA 4.0*
- 10.4: Working with Taylor Series - *CC BY-NC-SA 4.0*
  - 10.4E: Exercises for Section 10.4 - *CC BY-NC-SA 4.0*
- 10.R: Chapter 10 Review Exercises - *CC BY-NC-SA 4.0*

11: Parametric Equations and Polar Coordinates - *CC BY-NC-SA 4.0*
- 11.0: Prelude to Parametric Equations and Polar Coordinates - *CC BY-NC-SA 4.0*
- 11.1: Parametric Equations - *CC BY-NC-SA 4.0*
  - 11.1E: Exercises for Section 11.1 - *CC BY-NC-SA 4.0*
- 11.2: Calculus of Parametric Curves - *CC BY-NC-SA 4.0*
  - 11.2E: Exercises for Section 11.2 - *CC BY-NC-SA 4.0*
- 11.3: Polar Coordinates - *CC BY-NC-SA 4.0*
  - 11.3E: Exercises for Section 11.3 - *CC BY-NC-SA 4.0*
- 11.4: Area and Arc Length in Polar Coordinates - *CC BY-NC-SA 4.0*
  - 11.4E: Exercises for Section 11.4 - *CC BY-NC-SA 4.0*
- 11.5: Conic Sections - *CC BY-NC-SA 4.0*
  - 11.5E: Exercises for Section 11.5 - *CC BY-NC-SA 4.0*
- 11.R: Chapter 11 Review Exercises - *CC BY-NC-SA 4.0*

12: Vectors in Space - *CC BY-NC-SA 4.0*
- 12.0: Prelude to Vectors in Space - *CC BY-NC-SA 4.0*
- 12.1: Vectors in the Plane - *CC BY-NC-SA 4.0*
  - 12.1E: Exercises for Section 12.1 - *CC BY-NC-SA 4.0*
- 12.2: Vectors in Three Dimensions - *CC BY-NC-SA 4.0*
  - 12.2E: Exercises for Section 12.2 - *CC BY-NC-SA 4.0*
- 12.3: The Dot Product - *CC BY-NC-SA 4.0*
  - 12.3E: Exercises for Section 12.3 - *CC BY-NC-SA 4.0*
- 12.4: The Cross Product - *CC BY-NC-SA 4.0*
  - 12.4E: Exercises for Section 12.4 - *CC BY-NC-SA 4.0*
- 12.5: Equations of Lines and Planes in Space - *CC BY-NC-SA 4.0*
  - 12.5E: Exercises for Section 12.5 - *CC BY-NC-SA 4.0*
- 12.6: Quadric Surfaces - *CC BY-NC-SA 4.0*
  - 12.6E: Exercises for Section 12.6 - *CC BY-NC-SA 4.0*
- 12.7: Cylindrical and Spherical Coordinates - *CC BY-NC-SA 4.0*
  - 12.7E: Exercises for Section 12.7 - *CC BY-NC-SA 4.0*
- 12.R: Chapter 12 Review Exercises - *CC BY-NC-SA 4.0*

13: Vector-Valued Functions - *CC BY-NC-SA 4.0*
- 13.0: Prelude to Vector-Valued Functions - *CC BY-NC-SA 4.0*
- 13.1: Vector-Valued Functions and Space Curves - *CC BY-NC-SA 4.0*
  - 13.1E: Exercises for Section 13.1 - *CC BY-NC-SA 4.0*
- 13.2: Calculus of Vector-Valued Functions - *CC BY-NC-SA 4.0*
  - 13.2E: Exercises for Section 13.2 - *CC BY-NC-SA 4.0*
- 13.3: Arc Length and Curvature - *CC BY-NC-SA 4.0*
  - 13.3E: Exercises for Section 13.3 - *CC BY-NC-SA 4.0*

- 13.4: Motion in Space - *CC BY-NC-SA 4.0*
  - 13.4E: Exercises for Section 13.4 - *CC BY-NC-SA 4.0*
- 13.R: Chapter 13 Review Exercises - *CC BY-NC-SA 4.0*

14: Differentiation of Functions of Several Variables - *CC BY-NC-SA 4.0*
- 14.0: Prelude to Differentiation of Functions of Several Variables - *CC BY-NC-SA 4.0*
- 14.1: Functions of Several Variables - *CC BY-NC-SA 4.0*
  - 14.1E: Exercises for Section 14.1 - *CC BY-NC-SA 4.0*
- 14.2: Limits and Continuity - *CC BY-NC-SA 4.0*
  - 14.2E: Exercises for Section 14.2 - *CC BY-NC-SA 4.0*
- 14.3: Partial Derivatives - *CC BY-NC-SA 4.0*
  - 14.3E: Exercises for Section 14.3 - *CC BY-NC-SA 4.0*
- 14.4: Tangent Planes and Linear Approximations - *CC BY-NC-SA 4.0*
  - 14.4E: Exercises for Section 14.4 - *CC BY-NC-SA 4.0*
- 14.5: The Chain Rule for Multivariable Functions - *CC BY-NC-SA 4.0*
  - 14.5E: Exercises for Section 14.5 - *CC BY-NC-SA 4.0*
- 14.6: Directional Derivatives and the Gradient - *CC BY-NC-SA 4.0*
  - 14.6E: Exercises for Section 14.6 - *CC BY-NC-SA 4.0*
- 14.7: Maxima/Minima Problems - *CC BY-NC-SA 4.0*
  - 14.7E: Exercises for Section 14.7 - *CC BY-NC-SA 4.0*
- 14.8: Lagrange Multipliers - *CC BY-NC-SA 4.0*
  - 14.8E: Exercises for Section 14.8 - *CC BY-NC-SA 4.0*
- 14.R: Chapter 14 Review Exercises - *CC BY-NC-SA 4.0*

15: Multiple Integration - *CC BY-NC-SA 4.0*
- 15.0: Prelude to Multiple Integration - *CC BY-NC-SA 4.0*
- 15.1: Double Integrals over Rectangular Regions - *CC BY-NC-SA 4.0*
  - 15.1E: Exercises for Section 15.1 - *CC BY-NC-SA 4.0*
- 15.2: Double Integrals over General Regions - *CC BY-NC-SA 4.0*
  - 15.2E: Exercises for Section 15.2 - *CC BY-NC-SA 4.0*
- 15.3: Double Integrals in Polar Coordinates - *CC BY-NC-SA 4.0*
  - 15.3E: Exercises for Section 15.3 - *CC BY-NC-SA 4.0*
- 15.4: Triple Integrals - *CC BY-NC-SA 4.0*
  - 15.4E: Exercises for Section 15.4 - *CC BY-NC-SA 4.0*
- 15.5: Triple Integrals in Cylindrical and Spherical Coordinates - *CC BY-NC-SA 4.0*
  - 15.5E: Exercises for Section 15.5 - *CC BY-NC-SA 4.0*
- 15.6: Calculating Centers of Mass and Moments of Inertia - *CC BY-NC-SA 4.0*
  - 15.6E: Exercises for Section 15.6 - *CC BY-NC-SA 4.0*
- 15.7: Change of Variables in Multiple Integrals - *CC BY-NC-SA 4.0*
  - 15.7E: Exercises for Section 15.7 - *CC BY-NC-SA 4.0*
- 15.R: Chapter 15 Review Exercises - *CC BY-NC-SA 4.0*

16: Vector Calculus - *CC BY-NC-SA 4.0*
- 16.0: Prelude to Vector Calculus - *CC BY-NC-SA 4.0*
- 16.1: Vector Fields - *CC BY-NC-SA 4.0*
  - 16.1E: Exercises for Section 16.1 - *CC BY-NC-SA 4.0*
- 16.2: Line Integrals - *CC BY-NC-SA 4.0*
  - 16.2E: Exercises for Section 16.2 - *CC BY-NC-SA 4.0*
- 16.3: Conservative Vector Fields - *CC BY-NC-SA 4.0*
  - 16.3E: Exercises for Section 16.3 - *CC BY-NC-SA 4.0*
- 16.4: Green's Theorem - *CC BY-NC-SA 4.0*
  - 16.4E: Exercises for Section 16.4 - *CC BY-NC-SA 4.0*
- 16.5: Divergence and Curl - *CC BY-NC-SA 4.0*
  - 16.5E: Exercises for Section 16.5 - *CC BY-NC-SA 4.0*
- 16.6: Surface Integrals - *CC BY-NC-SA 4.0*
  - 16.6E: Exercises for Section 16.6 - *CC BY-NC-SA 4.0*

- 16.7: Stokes' Theorem - *CC BY-NC-SA 4.0*
  - 16.7E: Exercises for Section 16.7 - *CC BY-NC-SA 4.0*
- 16.8: The Divergence Theorem - *CC BY-NC-SA 4.0*
  - 16.8E: Exercises for Section 16.8 - *CC BY-NC-SA 4.0*
- 16.R: Chapter 16 Review Exercises - *CC BY-NC-SA 4.0*

17: Second-Order Differential Equations - *CC BY-NC-SA 4.0*

- 17.0: Prelude to Second-Order Differential Equations - *CC BY-NC-SA 4.0*
- 17.1: Second-Order Linear Equations - *CC BY-NC-SA 4.0*
  - 17.1E: Exercises for Section 17.1 - *CC BY-NC-SA 4.0*
- 17.2: Nonhomogeneous Linear Equations - *CC BY-NC-SA 4.0*
  - 17.2E: Exercises for Section 17.2 - *CC BY-NC-SA 4.0*
- 17.3: Applications of Second-Order Differential Equations - *CC BY-NC-SA 4.0*
  - 17.3E: Exercises for Section 17.3 - *CC BY-NC-SA 4.0*
- 17.4: Series Solutions of Differential Equations - *CC BY-NC-SA 4.0*
  - 17.4E: Exercises for Section 17.4 - *CC BY-NC-SA 4.0*
- 17.R: Chapter 17 Review Exercises - *CC BY-NC-SA 4.0*

Appendices - *CC BY-NC-SA 4.0*

- Appendix A: Table of Derivatives - *CC BY-NC-SA 4.0*
- Appendix B: Table of Integrals - *CC BY-NC-SA 4.0*

Back Matter - *CC BY-NC-SA 4.0*

- Index - *CC BY-NC-SA 4.0*
- Glossary - *CC BY-NC-SA 4.0*
- Detailed Licensing - *CC BY-NC-SA 4.0*